Independent Branches for Verification Serving

Anonymous research draft

Abstract

Formal verification services check related program and proof candidates that share libraries, declarations and accepted context. Reusing that preparation while serving independent attempts requires balancing worker isolation with resource growth, contention and cancellation. Native execution history includes scopes, heuristics, runtime objects and resource counters beyond logical context, so reconstruction needs a separate argument about observable behavior. Established caches, persistent environments, proof-state snapshots and bounded warm services recover preparation reuse, leaving unresolved the additional value of independent native execution at matched resources. We argue that reusable verification state should retain preparation while separating each candidate's execution history and lifetime from preceding alternatives. Realizing this separation requires supported quiescent boundaries, faithful native configuration and resource handling, and bounded memory as candidates modify branch-local state. We study a proposed verification-service design whose parent retains immutable prepared states and whose disposable candidate branches inherit native resource state, own mutable state and lifetime, and receive separate service accounting, with native execution for unsupported boundaries. We compare complete verification on public Lean and satisfiability modulo theories (SMT)-backed candidate streams against competent bounded warm services and native logical or serialized prepared-state reuse, then assess remaining heterogeneous checking after preparation reuse on an RTX 5090 with setup, transfer and CPU fallback charged. [**RESULT: End-to-end latency, throughput, memory and behavior results against the strongest matched services, including the measured scope of device benefit.**]

1 Introduction

Formal verification services check program and proof candidates against formal specifications. Related attempts from proof-generation systems share libraries, declarations and accepted context. Proof assistants construct and check proof terms, whereas program verifiers translate obligations into requests to satisfiability modulo theories (SMT) solvers. Community tools for Lean [4] serve these workloads through persistent environments [13].

Independent attempts require preparation reuse without coupling their lifetimes. Concurrent candidates need the same imported Lean context, but each must own the work that can be cancelled. Starting a verifier for every candidate repeats imports and parsing and replicates resident objects. Retained workers and shared logical environments avoid this repetition, while worker isolation determines resource growth, contention and cancellation behavior. Cancelling one attempt must release its speculative work without ending the others.

Native execution history extends beyond visible formulas and declarations. In Z3's combined solver, an additional scope can change the selected solving path [52], while earlier requests can alter heuristics and resource counters. Proof assistants also retain elaborator state and mutable runtime extensions beyond the immutable declaration environment [4, 13]. Logical reconstruction therefore needs a separate argument about observable verification behavior.

Established reuse recovers preparation, but the additional value of independent native execution at matched resources remains unresolved. Result caches avoid repeating completed work [24], and the community read–eval–print loop (REPL) provides persistent environments, backtracking and pickling [13]. Lean snapshotting retains elaborated proof state across parallel tactic branches [46], while stock Lean saves incremental elaboration snapshots between invocations [15]. Bounded warm services reuse environment roots and retire workers [41, 43], and Axle serves preloaded Lean contexts in separate request processes [50]. The comparison must establish whether separating preparation from candidate execution history improves complete verification, cancellation and total memory across service processes.

We argue that reusable verification state should retain preparation while separating each candidate's execution history and lifetime. Context prepared before candidate evaluation can serve independent attempts. Each candidate inherits that prepared native state, so preceding alternatives neither supply its solver history nor extend its lifetime. The benefit depends on preparation and branch-creation costs and on the pages each candidate modifies.

Realizing this separation requires runtime safety, faithful resource handling and bounded memory. Each branch must begin at a quiescent boundary that is safe for the verifier runtime. Native configuration and resource accounting must remain faithful when process creation changes timers or descriptors [27]. Prepared-state retention must also improve capacity within its

memory budget. These requirements determine the service's achievable combinations of latency, throughput and memory use at matched CPU capacity and memory limits.

The design we study retains immutable prepared states in a parent and runs candidates in disposable independent branches. Each branch inherits native resource state and owns its responses, mutable state, communication and cancellation. Separate service accounting records its costs. Supported quiescent boundaries, untouched candidate operations after preparation and bounded prepared-state lifetimes address runtime compatibility and memory use, with native execution for unsupported boundaries. We compare complete verification on public Lean and SMT-backed candidate streams against competent bounded warm services and native logical or serialized prepared-state reuse. After preparation reuse, we assess remaining heterogeneous checking on an RTX 5090, charging setup, transfer and CPU fallback to complete verification. **[RESULT: Matched behavior results and end-to-end latency/throughput/memory findings, including the measured scope of device benefit.]**

The evaluation tests four intended contributions. Section 6 distinguishes completed evidence from missing results.

- A characterization of sharing expensive preparation in actual ordered verification attempts (Section 6.2).
- A native branching design and an argument about observable behavior separating preparation from candidate interference (Section 4).
- A bounded implementation integrating independent branches with native verification and cancellation (Section 5).
- An equal-resource evaluation against established reuse, with a causal assessment of heterogeneous checking after preparation reuse (Section 6).

2 Background

2.1 Verification stages and outcomes

Lean source verification includes parsing, elaboration, proof construction and kernel checking [4]. Imports prepare declaration environments before a candidate is elaborated, while kernel checking validates proof terms against the dependent type theory [4]. By contrast, post-export checker replay begins after the source frontend has already run. These boundaries distinguish savings in candidate construction, library setup and proof checking [16, 13].

SMT-backed verification includes frontend translation, solver setup, parsing, preprocessing and solving [12, 52]. A native SMT session can contain multiple queries and output requests. Solver responses include SAT for satisfiable queries, UNSAT for unsatisfiable queries, and UNKNOWN when the solver does not decide the query. Clients can also request models, proofs or unsatisfiable cores when the configured backend supports them [52]. These outcomes and artifacts define the comparison in Section 4.2.

An accepted Lean command may admit an unproved obligation using `sorry` [13]. Complete proof verification distinguishes unfinished proof obligations and admissions from frontend diagnostics.

Closing a tactic's goals is also distinct from checking the enclosing modified declaration. LeanPolish tests tactic proposals against saved proof states before re-elaborating selected source edits [1]. Its native menu screen restores state without separately finalizing each alternative as the original declaration [22]. A successful tactic screen therefore establishes a different outcome from acceptance of the enclosing declaration or source.

2.2 Prepared native state

A native execution prefix is an ordered sequence of verifier operations together with its configuration and runtime state. For SMT, this includes options, declarations, assertions and scopes in their observed order [52]. For Lean, it includes imports and accepted commands that establish the environment or proof state before the next attempt. Persistent tools provide ways to retain some of this state [13, 46].

Native interfaces retain verification state, while runtime and resource properties determine how that state can be reused. The solver's own command interface saves state across evaluations [52], and a resource limit can bound internal work independently of elapsed time [53]. Process branching shares existing memory through copy-on-write, but child timers, descriptors and runtime threads need separate handling [27]. Section 4 uses these native-state and process properties to define preparation boundaries.

3 Motivation

3.1 Preparation and candidate lifetime

A service must account for preparation reuse and candidate lifetimes. Preparation has value when attempts reuse expensive state, rather than merely share similar text. **[RESULT: Final characterization of preparation CPU cost, memory and candidate fan-out from real Lean and SMT-backed streams.]** A service must also account for attempts that are rejected, remain incomplete or exhaust their resources. Their lifetimes consume capacity even when the client seeks only one accepted candidate.

Established persistent sessions make cold process execution an insufficient performance baseline. The commu-

nity REPL already allows multiple commands to refer to an earlier environment [13], and native proof-state snapshots support parallel tactic exploration [46]. We ask whether sharing native preparation changes concurrency, cancellation and aggregate memory relative to bounded versions of these services.

3.2 Scope and history effects

Logical equality of visible assertions does not establish equality of the native state. The distinction between logical context and execution history is observable in resource-bounded verification. [**RESULT: Final native-session and prepared-branch comparisons, including scope-only and prior-history controls for status differences.**] Scope effects can select a different solver path even in a fresh process [52], while history effects can arise without a change to the candidate's explicit assertions. A compatibility argument must distinguish both causes from timing variability and account for native resource and runtime properties when establishing a faithful branch boundary.

The preparation/isolation tradeoff and scope/history distinction motivate the design in Section 4. Preparation sharing, candidate isolation and bounded resource use are evaluated together. A speedup from a path that changes the verifier responses or requested artifacts is excluded from the service comparison, even when that path remains logically sound on some inputs.

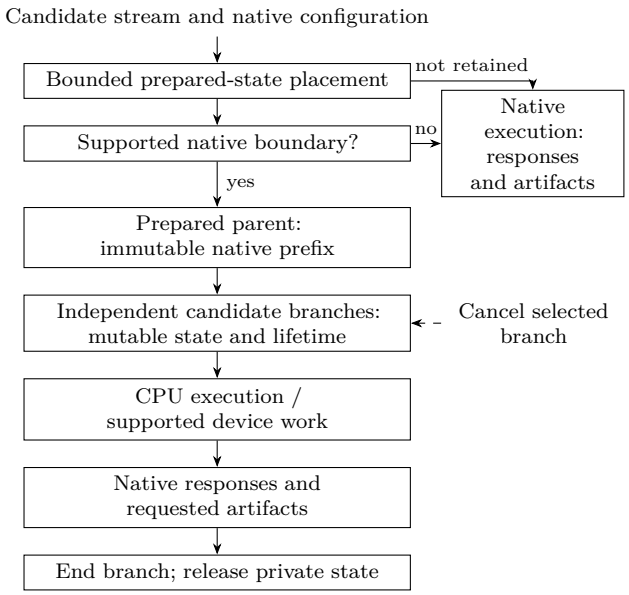


Figure 1: Supported branches inherit an immutable prepared prefix and own candidate mutation and lifetime. The parent performs no candidate verification after the selected boundary. Cancelling a selected branch ends its private work while leaving the parent and other branches independent. Normal completion releases private state after returning native responses. Unsupported boundaries use native execution. Placement can also select native execution without retaining a prepared state.

4 Design

The design has three goals derived from Section 3: (1) preserve the native verifier's observable behavior, (2) share expensive preparation while isolating candidate lifetimes, and (3) improve latency and throughput within matched CPU and memory resources. The service treats a prepared state as an immutable branching point. Candidate evaluation occurs only in an independent branch, and unsupported boundaries use native execution.

Placement first chooses whether retaining preparation is worthwhile, and requests without a retained state use native execution. For a selected prepared state, boundary qualification separately determines whether branching is supported (Figure 1). The parent owns the immutable prepared state, while each supported branch owns candidate mutation and its lifetime. Supported branches execute on CPU with supported device operations where available, while unsupported boundaries use native execution.

4.1 Preparation boundaries

A shared parent must represent state that every assigned candidate has available. The service therefore matches the original ordered preparation and configuration, without replacing them with a set of assertions. A candidate suffix begins after this preparation and includes its original scopes and output requests. The parent retains accepted context in the original preparation, but contains no learned or elaborated state produced by candidate suffixes after the selected boundary.

The boundary must also be safe for the runtime. Each adapter exposes a supported quiescent point with no pending candidate work or unresolved external effects. The service retains the prepared objects and initializes branch-owned communication and resource controls before evaluation. A runtime without such a point follows the native path. This restriction is a deployment boundary, not evidence that arbitrary native processes can be cloned safely.

4.2 Observable behavior

Native behavior is defined at the verifier interface, independently of the branch implementation. The reference is original native execution of the same ordered preparation and candidate suffix. This preparation retains the candidate's accepted context without inserting suffixes from preceding alternative candidates. History within the candidate's own native session remains part of its suffix.

Behavior comparisons cover verifier outcomes and requested artifacts. For Lean, the comparison includes di-

agnostics, completed proof obligations and admissions in each attempt. For SMT, it includes each SAT, UNSAT or UNKNOWN response and each requested artifact. Native proof or model validation checks artifact meaning where alternative valid artifacts can differ in representation. Counting accepted commands alone does not determine fidelity between verification paths.

The candidate executes its original suffix once in a branch. The service inserts no additional logical scope and performs no preparatory solver check to warm heuristics. Native solver resource state is inherited from preparation, while service CPU and elapsed costs are charged separately. These rules remove cross-candidate history as a source of difference, but wall-time limits and changed runtime conditions still require measurement rather than a universal assertion that responses remain equal.

The service chooses fallback according to interface compatibility, rather than treating every UNKNOWN response as a detector of divergence. Returning UNSAT when the native resource-bounded path returns UNKNOWN can also change client behavior. The service selects native execution when the adapter cannot support the boundary or requested artifact. Any residual mismatch in a supported comparison invalidates that path's fidelity claim. The service retains the mismatching case with its complete input and output.

4.3 Branch lifetime and cancellation

Candidate mutation should not force the service to retain the shared parent beyond its configured policy or affect another candidate. Each branch owns mutable candidate state and terminates after producing its response, while cancellation stops that branch and releases its private memory. Device work and buffers remain attributable to their candidate until completion or cancellation cleanup, which is included in branch lifetime and service accounting. The prepared parent remains available only while it satisfies the configured retention policy.

The lifecycle separates retained preparation from speculative work. A cancelled or incomplete attempt cannot become a new prepared parent. Branch failure returns an explicit native or service failure and leaves other branches independent. The evaluation includes failed attempts and cancellation recovery rather than reporting only successful candidate throughput.

4.4 Bounded placement and execution

The benefit of preparation sharing must offset the costs of retained memory and branch mutation. The service admits prepared states within a memory budget and accounts for proportional shared memory and private dirty pages. Proportional set size (PSS) attributes each shared page across the processes sharing it [26], and fleet memory sums these attributed amounts across service processes. Placement favors states whose observed preparation work is reused often enough to offset the costs of creating and retaining those states. Branch concurrency is limited by CPU capacity and memory pressure, so copy-on-write sharing does not justify unlimited fanout.

If candidates modify most retained pages, copy-on-write sharing can approach full replication. If preparation is cheap or locality is low, native execution avoids retention overhead. These cases test the same service tradeoff as workloads with substantial preparation sharing and define its measured operating range.

4.5 Heterogeneous checking

When preparation reuse removes repeated work, it leaves a smaller verification path for an accelerator to improve. The service selects expensive supported operations from profiles of that path and compares them with the strongest CPU implementation. For these checking operations, the input expression graph alone does not define dependencies, because reduction depends on binder environments and instantiated constants [4, 18]. A GPU operation therefore includes its actual dynamic work, required representations and synchronization.

Supported device work preserves verifier semantics. Before a device operation is admitted, device qualification compares its outputs with matched CPU references. During execution, enclosing native verification checks the resulting proof or requested artifact where one is produced, and required online checks are charged to complete verification. Qualification does not imply a second full CPU replay on every request. Response and artifact fidelity follow Section 4.2, including UNKNOWN and responses without artifacts.

Unsupported work returns to CPU execution, whose cost remains in the complete verification measurement. Complete verification includes device initialization, representation construction, upload, download and queueing. We report operation speedups measured with data already on the device separately from complete checking.

5 Implementation

The artifact implements the native adapters and branch constructors used to test the design. It contains separate experimental process-branch and shared-runtime state-branch constructors. These components do not yet establish the complete bounded service proposed in Section 4. Completed branch constructors have not demonstrated an improved equal-resource service frontier. Their behavior and performance results apply to the boundary and runtime version measured.

The SMT adapter retains Z3's native command context across native parser invocations to evaluate ordered session state [52]. The Lean adapter uses the runtime

boundary that prepares imports and accepted proof context before a candidate. For each verifier, qualification first compares fresh adapter execution with the original interface and then compares prepared branches with that reference. We measure reuse after these comparisons.

5.1 Native adapters

The native SMT command adapter is implemented in C++. It retains the command-line interface's command context and preserves the native order of configuration, manager creation and solver creation while evaluating original command sequences. It preserves options, scopes, checks and requested artifacts rather than reconstructing assertions through a different solver API.

Qualification compares fresh adapter execution with the matched standalone executable, using an unmodified executable built from the same source to distinguish build metadata from other response differences. This comparison separates interface incompatibility from differences caused by branching. Every observed response difference remains visible. Diagnostic branching tests can investigate a rejected adapter, but the service executes prepared-state branches only with an adapter that passes interface qualification.

A preparation boundary can separate parser invocations while retaining the native command context. Qualification therefore compares complete input, split native parsing and branch evaluation to distinguish parser end-of-input effects from branch effects.

For source verification backed by this SMT adapter, the measured interval also includes the native frontend's translation and its original solver interaction. Solver-session replay alone is reported at its narrower boundary.

For Lean, the adapter retains accepted context and keeps branch-local proof construction independent. It waits for source snapshots and the checked declaration environment before qualifying runtime quiescence, thread ownership and mutable extensions. Frontend completion alone does not establish that runtime workers, their finalizers and event loops have stopped. The adapter enables native process branching only after that separate runtime qualification.

5.2 Frozen arithmetic certificate baseline

The external native arithmetic oracle is retained as a frozen comparison baseline, not a contribution. External certificate search with native checking and floating-point support search followed by exact repair are established techniques; this composition does not supply systems novelty.

The arithmetic experiment tests whether reusing an external coefficient certificate removes work that remains after native compilation and warm environment reuse. FarkasFast selects hypotheses numerically and then invokes stock arithmetic on the selected subset [31]. That restricted invocation can repeat proof-producing normalization and certificate search. Its source does not establish a fixed count of completed oracle invocations. Our experimental path instead returns the numerical certificate and constructs only the facts that it cites.

For each contradiction branch, the Lean frontend collects the applicable assumptions and exports an arithmetic expression tree, an abstract syntax tree (AST), with stable identifiers for opaque atoms. A native untrusted process expands rational polynomials, generates nonlinear products and square facts, and performs support search with exact certificate repair. It returns nonnegative coefficients, equality orientations and the provenance of any selected product or square. Lean constructs those selected facts, forms their weighted sum and uses its native arithmetic discharger to check the resulting contradiction. The enclosing declaration still undergoes ordinary native checking.

One certificate check here means one weighted contradiction per solved branch, not one check for an entire tactic or source. For example, an equality goal can require two contradiction branches. The external search does not return trusted Lean expressions or replace the kernel. Unsupported syntax, a missed certificate or failed reconstruction restores the saved tactic state and selects the same Fast/hybrid and product-reuse path as the comparator. All failed attempts and fallback work remain in complete source-verification timing.

The certificate exporter in this composition directly handles real and rational arithmetic. Natural and integer arithmetic and unsupported normalization remain on the fallback path, so supported invocation counts alone cannot establish useful cost coverage. Literal-power handling in this direct path is not a strict type-coverage superset of the upstream numeric probe. This prototype uses CPU native code. Batching products on an RTX 5090 remains conditional on measured eligible work and complete preparation, transfer and fallback costs. External arithmetic search and native certificate checking are established techniques [31, 14], so their presence alone does not establish systems novelty.

5.3 Shared-runtime native state

The shared-runtime Lean constructor retains the prepared native command state while assigning each candidate a separate mutable-reference context. Reads of prepared reference cells use the retained base until a destructive operation creates a private cell. Unresolved prepared thunks are evaluated privately. Branch identity follows admitted native tasks, and branch release waits for those tasks and their execution scopes before releasing retained identities and private versions.

This reference overlay does not make arbitrary native globals, extensions, external effects or foreign-function interfaces (FFI) private. Listed guarded effects select native replay. When unsupported effects

terminate a worker, the operational outcomes remain recorded. These mechanisms establish neither general process equivalence nor completed cancellation qualification. The process constructor in Figure 1 and this runtime constructor have different supported boundaries and isolation properties.

5.4 Process and resource handling

Linux copy-on-write branches retain the parent address space, with candidate writes creating private pages [27]. The parent stays quiescent and never executes candidate suffixes, while branch communication uses separate descriptors. The service reaps children after response, cancellation or failure and distinguishes native solver resource counters from service elapsed and CPU accounting.

Cancellation instrumentation records the native progress reached before termination and whether the child was reaped before later work. The records show reaping after native progress. Establishing interruption inside a check requires separate execution-phase records.

Qualification of supported boundaries covers native resource controls affected by process creation. Claims about wall-time outcomes are limited to the measured operational scope described in Section 4.2 and the evaluation limitations, rather than following from inherited counters alone.

The design requires bounded prepared-state retention and branch concurrency using the preparation-reuse criterion in Section 4.4. Prototype instrumentation records parent and child proportional memory, private dirty memory and process lifetimes. These counters alone do not establish the proposed placement policy or a complete service frontier. This accounting distinguishes shared preparation from full worker replication and includes replacement costs.

5.5 Device path

The device path uses an allocated RTX 5090 through NVIDIA's container runtime [39]. During device qualification, a CPU reference and GPU implementation consume the same identified input and preserve output identity for each supported checking operation. Required online checks follow Section 4.5. Native artifact fidelity follows Section 4.2, and complete verification includes CPU fallback and device/runtime costs. For checker representation, existing techniques include expression layouts with integer identifiers and delayed substitution [18, 17], so their use does not establish a new checking algorithm.

The Linux implementation integrates native SMT session evaluation in C++, Lean context reuse and matched CPU/device execution through these adapters.

6 Evaluation

The evaluation examines four research questions, each mapped to the design goals and contribution it tests. RQ1 tests the sharing opportunity characterized by the first contribution. RQ2 tests behavior preservation and isolation, the first two design goals, while RQ3 tests the third goal, the equal-resource service frontier. RQ4 tests whether heterogeneous execution improves that frontier for complete verification after reuse.

RQ1 How much expensive native preparation can real verification candidate streams share across independent jobs?

RQ2 Does branching from prepared native state preserve verifier responses and requested artifacts while isolating candidates and cancellation?

RQ3 How does prepared-state branching change end-to-end latency, throughput and memory at equal resources compared with established warm-session reuse?

RQ4 After preparation reuse, when does heterogeneous execution improve complete verification over the strongest measured CPU path?

6.1 Experimental setup

We use complete ordered public proof and program attempts from multiple contexts in Lean and SMT-backed systems verification. Published proofs and deliberate invalid or incomplete variants provide behavior controls, while failed generated attempts remain part of the candidate-stream evaluation. These controls do not establish locality in generated candidate traffic. VeruSAGE provides real repository-level tasks [32, 51] for Verus [12], and Mathlib and the Lean Kernel Arena provide source/library and checker workloads [14, 16]. [**RESULT: Final source-native candidate-stream identities, versions, complete coverage and exclusions.**]

The SMT comparison replays 3,566 retained Rust candidates from the released AutoVerus artifact [33] in source-sorted order. Saved outputs include intermediate and unsuccessful attempts. This order does not reconstruct historical verifier arrivals.

Retained generated candidates and submission policy. The retained generated-candidate workload groups alternative proofs by task. The Goedel-Prover-SFT generation [6] contains 32 candidates for each of 244 miniF2F test tasks [54] and 186 ProofNet-lineage test tasks [42, 7], for a total of 13,760 candidates. Every selected candidate maps to its original task, model output, extracted source and submission/response events in four RTX 5090 generation segments. Interrupted tails and 24 additional returned responses outside the complete

32-candidate groups remain retained separately. The adapted SGLang [45] generator executes one request at a time. Official Goedel inference instead batches model inputs through vLLM [6, 49]. These event clocks record candidate availability, rather than verifier-request arrivals. The official next stage reads the completed code file and submits all candidates as a batch [5]. Consequently, finite bulk verification and online offered-load replay have distinct submission policies. Our earlier whole-source CPU comparisons load all inputs before service startup and use four outstanding HTTP requests. They are not exact reproductions of the official default 64-request scheduler.

Version-aligned released proof workload. We also characterize all 29,750 released GoedelWorkbook proof files migrated to Lean 4.27.0 and Mathlib revision a3a10db0 [34]. The retained metadata has 29,750 distinct problem identifiers, with one source per identifier. This cohort does not establish repeated same-problem preparation savings. We retain every published source file, including the publisher's migration errors and timeouts, in filename order. This release contains proof outputs, rather than every attempted generation or observed service arrival. Its competition-mathematics focus limits generalization to other Lean use.

One complete native-compilation reference acquisition uses four outstanding HTTP requests, shared CPU eligibility 0–3 and the pinned Kimina service with a 300-second request timeout and 12 GiB per-process address-space allowance. Candidate requests use its garbage-collection (GC) mode, which does not retain their command snapshots. It returns 27,319 complete, 2,376 rejected and 55 operational outcomes. The operational group contains 41 timeouts and 14 decoder failures. An individual native allocation-failure replay does not establish the cause of every decoder failure.

The main service competitors are bounded warm services with their existing cache and environment mechanisms, and native logical or serialized prepared-state reuse [41, 13, 15]. Existing REPL environment backtracking and native logical proof-state branching remain baseline implementations. Fresh native execution supplies the behavior reference and a preparation control. Matched frontier comparisons require the same candidate information, CPU capacity, memory budget and admission order. The earlier standalone GPU arithmetic service uses additional server CPU resources, so its measurements do not establish that frontier. The later additive-portfolio qualification matches CPU eligibility with its CPU-only control, but a short qualification does not establish the complete service frontier.

For the matched Lean 4.27 native-batch comparison, the primary competitor is the existing Kimina naive `BatchCommand` path [41], which runs native tasks from one prepared root. Disabling mutable-reference branching in the candidate method is a separate component control. All methods return an ordered batch array after its last member, so batch emission is not compared with streaming completion as a latency advantage.

The earlier eager-copy whole-corpus acquisition is stopped with its complete control and partial candidate data retained. The subsequent first-write runtime has finite native engagement and matched diagnostic checks, but no stable advantage over stock. This runtime constructor is closed. Neither version provides a completed comparative performance result. The native-batch experiment inherits the hosting environment's address-space policy after all methods fail under the service's 12 GiB policy. It therefore does not establish equivalence to the memory policy of a default Kimina deployment.

The heterogeneous comparison uses the fastest runnable complete CPU checker at a matched boundary, in addition to official Lean. A kernel checker operates after export only when the measurement explicitly uses that boundary. **[RESULT: Pinned strongest service artifacts and matched configuration.]**

We preserve all raw per-input outcomes. Lean outcomes include unfinished proof obligations and admissions as well as frontend diagnostics. A successful tactic screen requires the enclosing declaration or source checking stage before it counts as an accepted proof.

Primary metrics are end-to-end candidate latency distributions, completed attempts per second, corpus makespan, CPU time and fleet proportional memory. Queueing, preparation, parsing, elaboration, proof construction, checking, transport, setup and teardown are included where they occur. Throughput cost per completed job is distinguished from request latency.

We repeat whole comparisons with alternating condition order. Affinity specifies permitted CPUs but alone does not provide CPU isolation [29], so we record affinity with topology and observations of competing work. **[RESULT: Hardware, topology, software versions, repetitions, observed load, uncertainty and measurement details.]**

For the bounded arithmetic comparison, the host is an Intel Core Ultra 9 285K with 24 physical cores and no SMT siblings in the observed topology. Both arithmetic frontends use Lean's existing integer arithmetic tactic omega [20]. Earlier arithmetic clients use CPUs 12–15. The complete instrumented common-omega and natural-power blocks use 4–7; ready-goal qualifications use 8–11, and the retained prefix captures use 20–23. Affinity does not reserve memory bandwidth or isolate other work. The arithmetic runtime is Lean 4.27.0 with Mathlib a3a10db0, and the robust support controls use HiGHS 1.15.1 and cuOpt 26.10.0 with CUDA 12.9 [10, 38]. The RTX 5090 is shared through device-plugin time slicing with another Workspace, without a fixed half-device allocation. The new full-population arithmetic protocol uses two source requests and at most two live REPLs, including warm, starting and retiring processes. Each REPL has a 24 GiB address-space limit. An independent par-

ent samples total owned-process resident set size (RSS) every second against a 48 GiB budget and records any largest-REPL termination as a resource failure. This sampled response is not a kernel-enforced RSS ceiling. All methods share four eligible CPUs 8–11. The real 24-source preflight has completed.

The earlier external cuOpt server can execute on CPUs 0–23 while clients use 12–15, so that follow-up does not establish the equal-resource frontier of RQ3. Its client CPU and memory observations also exclude the external server. Server counters include the qualification window and do not provide exact per-cell accounting.

6.2 Preparation sharing

RQ1 asks how much expensive native preparation real verification candidate streams share across independent jobs. We preserve original attempt ordering and measure preparation CPU time and resident state available before each attempt. We separately analyze result reuse for exact duplicate attempts, sharing of ordered native prefixes and sharing of native environments. Sharing a native environment characterizes established reuse, while a supported process branch must also satisfy the preparation and boundary conditions in Section 4.1. The characterization includes failed and incomplete attempts, so locality is measured across all attempts rather than only successful proofs.

A complete source-level characterization finds one original task header in the miniF2F pool and 17 in the ProofNet-lineage pool. Apart from extraction-failure strings, the pools contain zero and 121 repeated code occurrences, respectively. Header recurrence and source duplication do not measure native preparation CPU or memory. In the complete 29,750-source modern GC captures, the original service executes 67 import headers and selects 29,683 started-REPL reuses. The compiled route executes 60 headers and selects 29,690 reuses. All source IDs are accounted for once. These counts confirm that the competent baseline already reuses preparation, but do not measure its CPU or latency share. Older preparation controls also include an explicit `LibrarySearch` query for `True`. Their initialization intervals therefore do not establish preparation demand derived from the candidate requests.

A separate request-driven observation measures all 5,952 ProofNet-lineage candidates on the producer-compatible Lean4.9 GC and selective native arithmetic service, without the injected search query. Stock header splitting maps 5,947 requests to one `import Mathlib` key and leaves five literal extraction-failure strings with empty headers. In each of three whole observations, the service performs five nonempty import preparations and 5,942 started-worker reuses. Preparation consumes 12.31, 12.99 and 12.75 observed native CPU seconds; the corresponding observed lifetime totals are 1,522.98, 1,531.77 and 1,521.46 CPU seconds. Their ratios are 0.81–0.85% of observed native CPU, not an exhaustive CPU fraction or an end-to-end speedup bound. Each observation returns 413 complete, 5,534 rejected, four incomplete and one operational outcome in 468.27, 470.37 and 465.68 seconds, respectively. The failed worker's unobserved CPU tail remains censored. Empty-header initialization can occur in its first body and is not claimed free. These observations do not establish consequential remaining import preparation cost after competent reuse on this cohort.

A phase reanalysis of the complete SMT streams measures the cost paid to prepare a reusable parent. Each branch observation prepares 473 parents across 136 prefix identities. Their cumulative CPU from process birth to readiness totals 7.568, 7.636 and 7.712 seconds, with per-parent medians of 14.9, 14.8 and 15.2 milliseconds. Logical reuse totals 7.699, 7.571 and 7.537 seconds under the corresponding three primary observations. These counters include native initialization before parsing and exclude subsequent candidate suffixes, the Verus frontend, transport and service management. They are neither a whole-verification CPU fraction nor measured avoided work. In particular, 2,316 parent hits do not imply 2,316 avoided expensive preparations: logical reuse still reparses and translates each candidate. This modest absolute readiness cost supplies no reason to reopen the slower process-branch constructor in Section 6.4.

Shared tactic-prefix opportunity. We apply SHRED's native tactic parser [47] to all retained miniF2F and ProofNet-lineage candidates, preserving their original groups of 32. Reuse keys include the task and the original statement because generated candidates can contain different statement text for the same task. The parser admits 7,672 of 7,808 miniF2F candidates and 5,717 of 5,952 ProofNet-lineage candidates. The remaining 136 and 235 candidates require the original verification path because extraction or conservative native tactic splitting is unsupported.

Within the eligible candidates, exact rooted edges include comments and whitespace between tactics. Only 8 of 29,669 miniF2F tactic occurrences and 45 of 21,530 ProofNet-lineage occurrences repeat a preceding rooted edge, corresponding to 0.027% and 0.209%. Keys based on the parser's tactic-unit text omit inter-tactic trivia while retaining the tactic text itself. They expose 3,987 and 3,957 reusable occurrences, or 13.44% and 18.38%. These are occurrence counts rather than measured CPU coverage or time saved.

Existing closing-certificate reuse. We also integrate SHRED's unchanged closing-certificate module and wrapper with the official Kimina HTTP application [47]. On all 5,952 original ProofNet candidates, native verification takes 356.37 seconds and this integration takes 373.22 seconds, including actual input preparation and process retirement: 4.73% more elapsed time

in one fixed-order shared-host pair. Both methods genuinely complete the same 413 sources. Native outcomes are 5,534 rejected, four incomplete and one operational failure; reuse returns 5,514 rejected, 24 incomplete and one operational failure. The twenty changed sources carry cache-hit messages and sorry warnings. The software development kit (SDK) reports valid counts of 417 and 437, which are therefore not complete-proof counts. Opaque environment handles and messages also differ. Eight outstanding sources share four eligible CPUs and the same 48 GiB per-process limit. This is an as-integrated existing-method control, not a deployment of an upstream HTTP service supplied by SHRED or a matched transparent speedup. A separate complete miniF2F pair uses two live REPLs, each limited to 24 GiB, with a 48 GiB total owned-process RSS watchdog in both methods. All 7,808 original candidates take 18,987.74 seconds natively and 19,235.45 seconds with reuse: 1.30% more elapsed time. Genuine completions decrease from 3,015 to 3,010. Eleven native rejections instead return sorry-bearing incomplete results after cache hits; five native successes report a kernel unknown-constant error for `_auxLemma.1`, also after hits. The SDK's 3,021 valid reuse results therefore do not establish more complete proofs. Both methods retain the same 118 operational-failure IDs, with timeout/resource classifications of 71/47 and 68/50. Native outcomes are not preserved by this integration. This is one shared-host observation at the stated resource point, not a general rejection of prefix reuse. Figure 2 retains all source latencies.

Earlier parser-based execution comparisons appear in Appendix B. They retain changed outcomes and operational failures rather than supplying a validated speedup. [**RESULT: Complete operational adjudication, repeated observations, cost-weighted savings and claim-matched upstream workflow/service comparisons.**]

We compare original ordering with a control containing unrelated work and measure how preparation depth changes reusable work. Text bytes and declaration counts are descriptive inputs, while measured preparation cost determines the opportunity. [**RESULT: Preparation-cost distributions across remaining contexts, context reuse over time and aggregate achievable sharing with uncertainty.**] RQ1 remains unanswered until these measurements of complete original candidate streams establish whether preparation sharing is consequential.

6.3 Behavior and isolation

RQ2 asks whether branching from prepared native state preserves verifier responses and requested artifacts while isolating candidates and cancellation. We first compare fresh adapter execution with the original verifier, then execute untouched candidate suffixes from prepared states. We retain per-input decisions and artifacts for comparisons that vary preceding unrelated attempts, branch cancellation and resource-limit conditions. Completed attempts are compared for native responses and artifacts. Externally cancelled attempts are compared for progress at termination, cleanup and the independence of surviving attempts. Native scope and history controls distinguish changes in solver mode from cross-candidate interference.

Observed behavior within the completed service comparisons. The SMT comparisons in Section 6.4 preserve native inputs, basic results, decision sequences and requested artifacts on their reported primary and resource conditions. The ProofNet service replay preserves basic outcome counts, whereas the shared-process placement retains diagnostic differences in the numbers assigned to metavariables, which are placeholders used during elaboration, on incomplete sources. These comparisons use distinct constructors and comparison scopes. Basic outcome agreement does not establish identical response bytes or general artifact fidelity. Section 6.4 retains their timing and resource conditions. The remaining history, resource-limit and cancellation comparisons below still determine RQ2 at its full scope.

Lean controls cover complete acceptance, rejection, unresolved obligations and admissions. SMT controls cover SAT with model requests, UNSAT with supported proof/core requests, UNKNOWN and multi-check sessions. The original scope/history regression streams remain part of the causal test. [**RESULT: Complete per-input comparisons, artifact validation, resource-limit outcomes and cancellation recovery.**] RQ2 remains unanswered until every planned behavior comparison finishes with a recorded outcome, residual differences are explained and divergent paths remain excluded from qualified service comparisons.

6.4 Latency, throughput and memory

RQ3 asks how prepared-state branching changes end-to-end latency, throughput and memory at equal resources compared with established warm-session reuse. We replay actual attempt streams over offered-load and concurrency sweeps, matching CPU and memory budgets across services. A single shared warm worker is not the sole comparison to many branch workers. We charge the measured service for creating and replacing prepared states and for cache costs.

We report latency distributions and completed throughput together with fleet proportional memory and CPU work. The causal comparison includes startup-only branches, deeper prepared prefixes and independent warm workers. Ablations remove prefix preparation, vary preparation depth and replace placement with ordinary bounded retention. Private dirty pages and branch lifetime explain whether sharing or simple warm

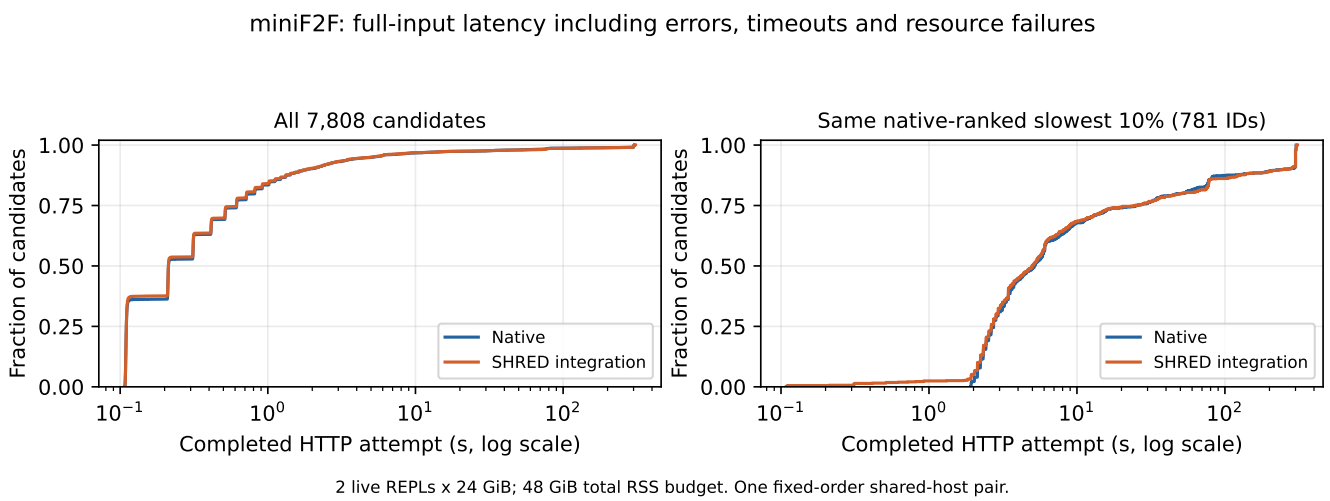


Figure 2: Complete miniF2F native/reuse HTTP latency: all 7,808 inputs per method, including errors, timeouts and resource failures (left); the same 781 native-ranked slowest inputs (right). Two live REPLs share four eligible CPUs and a total 48 GiB RSS budget. HTTP spans exclude SDK predispatch and global input preparation, which whole-flow clocks charge. This fixed-order, shared-host pair has changed proof outcomes as described in the text.

startup drives the observed gain. [**RESULT: Matched frontier plots, tail latency, memory, cancellation cost and mechanism ablations.**]

The comparisons below test distinct execution and placement mechanisms under their stated workloads and service conditions. Their observed improvements and losses apply to those constructors. They do not combine into one implemented branching system. The complete frontier comparison must also account for the full workload and resource matrix.

Single native-reference latency distribution. The GoedelWorkbook native-compilation reference uses the setup described above. Figure 3 separates these outcome populations. Completed proofs have median request latency 0.399 seconds, p95 3.324 seconds and a maximum of 272.692 seconds. These are closed-loop request durations, which overlap and do not decompose whole-corpus wall time or kernel CPU cost.

Complete ProofNet service replay. We replay all 5,952 retained ProofNet-lineage source attempts through a local native stream interface [42, 7]. The warm pool retains native workers across candidates, independent branches replace a worker after each uncached candidate, and snapshot/cache uses separately started native full-stream workers. All methods use the same success-only exact header/source cache policy, four worker slots and the same four permitted CPUs. They retain the Lean4.9.0-rc1/Mathlib environment, native runtime threads and original 300-second allowance from dispatch. Queue waiting does not consume that verifier allowance.

This retained protocol also explicitly warms a library-search index with a `True` query before candidate checking. Its observed wall interval is 6.69–6.93 seconds in the warm-pool and branch captures. The snapshot worker calls the same query without emitting a separate timer. These measurements therefore characterize this modified preparation protocol. They do not measure preparation demanded by the original requests or an untouched official service. No ablation isolates the query's effect on queueing or the method ratios.

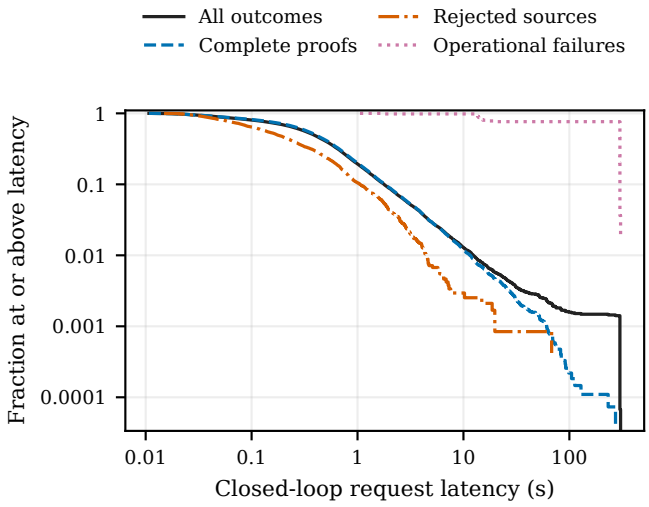


Figure 3: Request-latency survival curves for one complete 29,750-source native CPU reference acquisition. Each curve is normalized within its indicated outcome population. Both axes are logarithmic. Operational outcomes remain included. This single acquisition characterizes the workload tail and provides no between-run uncertainty, speedup or offered-load queueing result.

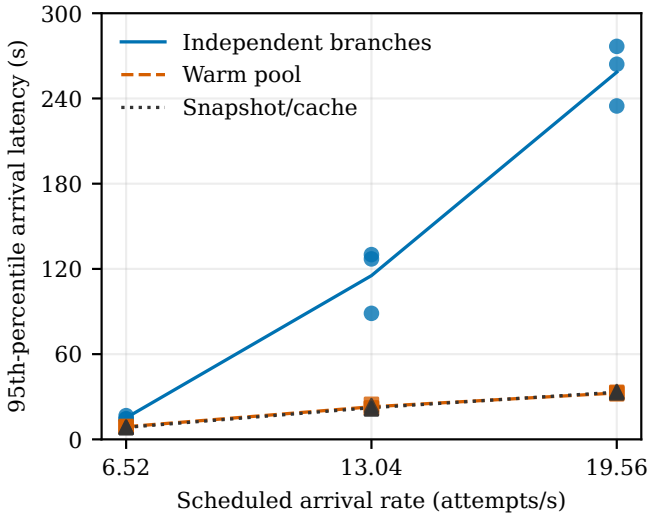


Figure 4: Arrival latency under synthetic load over 5,952 original ProofNet-lineage attempts per condition. Markers show all three whole-run p95 observations for each method and scheduled rate. Lines join their arithmetic means, not pooled request percentiles or confidence intervals. All 27 primary conditions complete on the same four permitted CPUs. Independent branches have worse p95 latency at each tested rate. The separate profiles are excluded. This finite replay does not establish a production arrival trace or steady-state capacity.

Each method receives the original source order at deterministic synthetic arrival rates of 6.52, 13.04 and 19.56 attempts per second. Method and rate orders rotate over three complete primary repetitions. Three separate resource profiles bring the complete ProofNet cohort to 30 conditions. Every condition returns 413 complete, 5,534 rejected, four incomplete and one operational outcome, matching the retained stock basic outcomes. Arrival latency starts at the scheduled arrival and ends at captured response, including release lag and pending-queue time. Whole-corpus clocks additionally include initialization, final drain and end of file (EOF).

Figure 4 rejects a claim of faster service for this per-candidate branch implementation at the tested preparation boundary. At the highest rate, the branch implementation's p95 is more than seven times the warm pool's in each repetition. Low-rate whole times mostly follow the arrival schedule, so their similarity does not establish equal service capacity. At the two higher rates, branch mean waiting before dispatch is 41.658–142.095 seconds, compared with 0.363–0.395 seconds after dispatch. Mean release lag stays below one millisecond. These means locate waiting within the measured implementation. They do not isolate process-cloning cost or identify CPU-time fractions.

Separate profiles observe sampled PSS maxima of 2,764,774 KiB for branches, 4,806,597 KiB for the warm pool and 8,857,235 KiB for snapshot/cache. The branch profile has incomplete observations in 187 of 633 samples. One-second non-atomic sampling misses short-lived processes and peaks, so these single profiles do not establish replicated memory savings or a complete memory frontier. The wider workload, context and resource comparisons remain open.

Native SMT prefix comparison. The native SMT prefix comparison rejects acceleration for the tested constructor and cohort. Three complete rotations check all 3,566 retained AutoVerus source candidates through stock native execution, prepared native branches and logical reuse. The matched conditions use the same three permitted CPUs and original native configuration, charging startup, transport, raw capture, archival and cleanup. Stock whole times are 1,561.1, 1,618.0 and 1,889.7 seconds. Branch times are 2,124.1, 2,231.5 and 2,434.8 seconds. Branches are 28.84–37.91% slower and use 22.67–25.05% more CPU in the recorded counter.

The recorded CPU counter combines the acquisition process clock with `getrusage` child totals, which depend on completed waits [28]. Exhaustive descendant coverage is not established, so the ratios do not establish total process-tree CPU overhead.

All 10,698 stock/branch source comparisons preserve native inputs, basic results, decision sequences and requested artifacts. Each branch condition engages 2,316 retained-parent hits and 473 misses, so the measured elapsed-time and recorded CPU increases occur with actual reuse. Branch/logical ordering varies, and logical-stream completeness annotations remain in two rotations. All three separate resource captures also complete, with matching native input, results and requested artifacts across each pairing.

Separate resource captures observe sampled PSS maxima of 224,644 KiB for stock, 783,477 KiB for branches and 791,914 KiB for logical reuse. Between 233 and 368 samples per method contain incomplete reads. These single non-atomic profiles do not establish replicated or exact memory bounds. They show no stock-relative memory saving that compensates for the measured time loss. The complete service frontier remains unfinished.

Complete source verification across placements. We compare a shared-process warm-worker alternative with the competent process-isolated warm pool on the retained 5,952 ProofNet-lineage source attempts [42, 7]. Both use the retained Lean environment and reuse native incremental snapshots, prepare the first imported environment and search index, and use four frontend workers on the same four permitted CPUs with default native runtime threads. The warm pool already shares its prepared roots through copy-on-write. The alternative shares one native heap and runtime while keeping each worker's root table separate as candidates update it. Complete clocks include initialization, preparation,

Whole corpus time (s)	Run 1	Run 2	Run 3
Process-isolated warm pool	303.029	303.826	303.671
Shared-process alternative	371.637	360.690	349.614

Table 1: Complete native source verification for the shared-process placement and warm pool, each with 5,952 original attempts. The three primary pairs retain identical basic outcomes. Resource profiles are separate conditions.

source verification, response transport, actual failure recovery, output, EOF and cleanup.

Table 1 rejects an acceleration claim for this shared-process placement on the measured cohort. The shared-process alternative is 22.64%, 18.72% and 15.13% slower than the warm pool. Each condition returns 413 complete, 5,534 rejected, four incomplete and one operational outcome. A native panic ends the shared process. The complete method checks every unreturned original candidate through the ordinary native warm path and charges both phases. The primary pairs recover 13, 14 and 13 candidates in this way. Internal metavariable numbers in diagnostic suggestions differ on two retained incomplete sources in some pairs. These responses remain unnormalized, so matched basic outcomes do not establish identical artifact bytes in general.

Separate profiles observe sampled fleet PSS maxima of 7,156,560 KiB for the shared-process alternative and 7,574,628 KiB for the warm pool. The 5.52% lower sampled PSS is a single comparison, not replicated memory evidence. One-second non-atomic sampling and competing mappings limit PSS interpretation. A shared heap also changes whether unchanged anonymous pages are counted as private, so private-page totals across these placements do not directly measure mutation.

Response extraction after warm preparation. An optimization of response extraction yields a small complete verification improvement on the same retained ProofNet cohort. The producer REPL classifies explicit `sorry` nodes by printing an entire syntax node and comparing its string [13]. We add a necessary check of the shortened parser-node kind before the unchanged full-string predicate. The optimization retains the native tree traversal, goal snapshots, diagnostic construction, enclosing frontend and kernel.

Three alternating complete primary pairs use the same warm-pool host, prepared state, four permitted CPUs and native defaults. Original whole times are 303.124, 308.703 and 301.252 seconds, compared with 297.520, 301.563 and 299.071 seconds with the precheck. The observed wall reductions are 1.85%, 2.31% and 0.72%, with a descriptive mean of 1.63%. This observed range is not a confidence interval.

Each pair preserves all 5,952 basic outcomes. Requested responses are identical in two pairs. One source in the other pair has internal metavariable-number differences in diagnostic hints, which remain unnormalized.

Separate profiles contain complete timer sequences in both methods for 5,791 source indices. Over this subset, cumulative calling-thread CPU falls from 8.476 to 0.664 seconds for sorry extraction and from 20.170 to 12.296 seconds for complete response extraction. Interleaved or incomplete records are censored. These region totals exclude asynchronous frontend threads and subsequent response serialization and construction of auxiliary declaration records. These totals are not fractions of corpus wall time.

Separate resource profiles observe sampled fleet PSS maxima of 7,860,678 KiB for the original method and 7,728,536 KiB with the precheck, which do not establish replicated memory savings. This optimization strengthens a CPU comparison on the measured cohort. It does not establish systems novelty or heterogeneous benefit.

First-success candidate scheduling. A distinct inference consumer needs one checked proof per problem rather than a reward for every candidate. In the completed native miniF2F capture, 142 of 244 problems have at least one valid candidate among their 32 attempts. Only 17 of the 83 timeouts belong to those solvable problems; 66 belong to problems with no observed successful candidate. Cancelling siblings after a checked success can therefore remove only part of the timeout cost.

An offline two-worker replay treats each observed HTTP span as a fixed service cost and assumes free prediction and cancellation. Relative to the same all-candidate replay, original-order and source-size first-success policies give modeled ratios of 1.45–1.48× while retaining the same 142 solved problem IDs. A hindsight order that knows the fastest valid candidate gives 1.50×. These are optimistic scheduling models, not measured speedups or physical upper bounds: changing order can change native costs, and actual setup, cancellation and retirement remain unpaid in this model. Source-size ordering also reduces solved-problem coverage at some intermediate candidate counts. Real paired execution is pending. First-success cancellation does not provide every candidate reward required by reinforcement-learning training, and it does not establish a new contribution for ordinary scheduling.

RQ3 remains unanswered until branching is compared with the strongest competent services over the complete workload, offered-load and resource matrix.

6.5 Heterogeneous execution

RQ4 asks when heterogeneous execution improves complete verification over the strongest measured CPU path after preparation reuse. We identify whether each complete measurement begins with source verification or post-export checking. Results from post-export checking therefore apply only to that checking boundary. We

profile warm complete checking and identify costly operations and their actual context-sensitive dependencies. Device qualification runs the same supported operation on CPU and RTX 5090 with exact reference comparisons, while required online checks follow Section 4.5. All unsupported work remains in the complete verification interval.

We separate host representation construction, device setup, transfers, checking with data already on the device, synchronization and fallback. The complete CPU/device comparison uses the service accounting and matched workload boundary of RQ3. An improvement in a checking operation alone therefore does not establish an improvement in the service frontier. The comparison includes tuned alternative CPU checkers with different parsing and reduction algorithms [18, 17]. **[RESULT: Cost coverage, amount of checking work simultaneously ready to execute, complete CPU/device comparisons and workload crossover.]**

Within the arithmetic CPU family, existing Farkas [31] provides native exact certificate search and numeric hypothesis selection before stock proof reconstruction. Our HiGHS support-search adapter uses the established linear programming (LP) solver [10], followed by exact certificate repair and Lean's standard proof construction and checking. These are CPU competitors in the heterogeneous comparison.

The CPU comparisons establish the reference for the device experiments. Here tiered denotes native exact certificate search, while hybrid denotes native double-precision (FP64) support search followed by exact repair. Fast/hybrid combines the upstream FarkasFast numeric probe and restricted stock proof construction with the native hybrid route and full native fallback [31].

6.5.1 Arithmetic comparisons across workloads

The selected-cohort representation and deployment controls appear in Appendix C.1. The comparisons below retain the complete original population and the version-aligned workload that limits generalization.

Complete original population. The new six-cell service comparison returns all 29,750 original sources in every cell at source concurrency two, at most two live REPLs, 24 GiB per REPL, 48 GiB aggregate sampled RSS and four CPUs 8–11. Table 2 reports all three alternating pairs. The ratio of summed whole times is 1.10627×, or 9.61% less elapsed. Preparation, imports, SDK waiting, fallback and process retirement are charged. Shared input materialization costs 0.330 seconds separately. The six-cell comparison over all 29,750 original sources is complete. Its observed whole-time ratio is 1.10627×, with two native rejection-to-completion changes per pair; the full outcomes and resource point are reported below. Its clocks remain separate from the earlier resource points. This complete population supports a modest CPU composition improvement; the selected heavy-cohort magnitude does not transfer to all originals.

Pair	Native (s)	CPU oracle (s)	Ratio
1	18,584.96	16,661.35	1.11545
2	18,478.26	16,912.11	1.09260
3	18,608.55	16,750.33	1.11094

Table 2: Complete current-resource Goedel population. Every cell returns 29,750 sources; orders are native–oracle, oracle–native and native–oracle. Two native rejections become oracle completions in each pair. Ratios are descriptive observations with changed tactic coverage, not transparent outcome equivalence or a GPU result.

Each native cell has 27,334 complete, 2,376 rejected and 40 timeout outcomes. The oracle has 27,336 complete, 2,374 rejected and the same 40 timeouts. The only changed source IDs, Proof_20758 and Proof_29379, have native `linarith` search failures but complete through the oracle path with native checking. Both released originals already record arithmetic-search failure. No source is removed, and no resource failure or sorry-bearing incomplete result occurs. These outcomes prevent an unchanged-output claim. Maximum sampled total RSS is 21.17 GiB, with at most two live REPLs, no watchdog kill or monitor error, and no owned descendant left at retirement.

For each candidate cohort, we sum HTTP request envelopes separately for the CPU oracle and native hybrid. Figure 5 divides the CPU sum by the native sum; values below one favor the oracle. The reciprocal ratios below use the same paired source IDs.

Within the same full cells, the original selected 1,000 IDs have overlapping HTTP-envelope-sum ratios of 1.75–1.83, while the remaining 28,750 have 1.06–1.08. This is a within-population comparison, not a new subset lifecycle measurement or a pooling of historical clocks. The same-native-ranked slowest 1% have envelope-sum ratios of only 1.024–1.038; the slowest 10% have 1.083–1.104. The 40 timeouts contribute about 12,002 overlapping HTTP seconds per cell. The oracle has longer envelopes on 4,106 IDs in all three pairs. These spans include queueing and overlap, so they neither identify removable CPU work nor establish a GPU Amdahl ceiling. Three shared-host observations are not a confidence interval. No GPU executes in this matrix. The separate native cost diagnostic below partitions observed arithmetic work, but it does not turn these source spans into a scheduler-independent GPU ceiling.

Version-aligned ProofNet generalization. A separate version-aligned component comparison processes every original ProofNet candidate in Lean 4.9.0-rc1. Module-loaded stock fallback, native preprocessing with

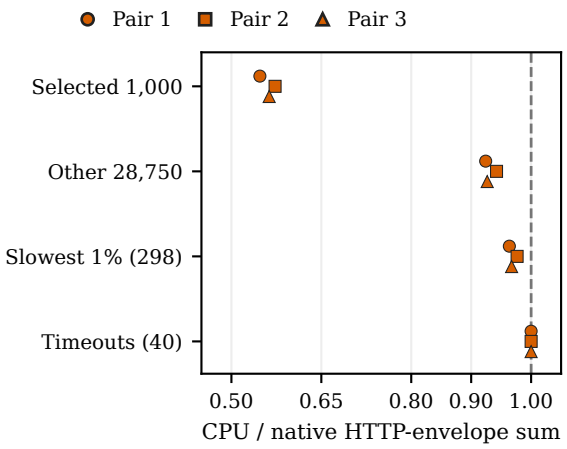


Figure 5: CPU/native ratios of summed HTTP request envelopes within source cohorts from the three complete 29,750-source pairs. Symbols identify pairs; cohort rows overlap. The slowest 1% is selected by each pair's native times. The horizontal axis is zoomed around parity; lower is faster. The selected cohort benefits most, while timeouts and the hardest tail barely change. Envelopes overlap in wall time, so their sums are neither complete elapsed time nor removable CPU work. No GPU executes in these cells.

external search, and direct certificate checking take 1,151.88, 1,160.92 and 1,178.93 seconds, respectively, in one fixed-order shared-host block. All three retain 413 complete, 5,523 rejected, 12 operational and four incomplete outcomes, with matching per-source verdicts and non-message fields; four and five warning differences remain in the native and direct modes. Direct execution is 2.35% slower than the module-loaded control, so the selected Goedel result does not generalize to this ProofNet component. The native mode is not the full Lean 4.27 Fast frontend. The stock control imports the same adapter, whose explicit argument elaboration precedes mode rejection. It is therefore not pristine Mathlib. All three failed REPL requests per condition and their retirements remain charged. This is neither a whole-service comparison nor GPU evidence; the complete miniF2F arithmetic comparison remains pending.

6.5.2 Arithmetic portfolio and device cost

Current native search attribution. A complete diagnostic over the original 1,000 heavy candidates uses two live REPLs, 24 GiB per REPL, a 48 GiB aggregate sampled RSS budget and four CPUs 8–11. Its own complete lifecycle is 1,380.32 seconds. All 992 complete and eight rejected responses match the frozen native reference in full. The 8,646 native LP requests attributed to source IDs contain 39.94 accumulated wall seconds of floating-point solve and 951.85 seconds in 1,636 exact fallbacks. Positive support repair totals 0.60 seconds. Exact fallback includes matrix construction, coefficient conversion and rational cleanup. Nonlinear product generation totals 225.61 seconds. These concurrent phase sums are not disjoint fractions of whole elapsed time.

Actual packed matrices have median 37 rows, 136 columns and 727 nonzeros; the maxima are 888, 3,160 and 26,044. The trace contains 3,102 cross-source matrix-ready-to-answer overlap pairs, with at most two open requests. These are hypothetical race windows, not GPU batches. Among the sources ranked slowest on the native reference, the slowest 1% includes only nine exact fallbacks, but they occupy 108.36 accumulated seconds. This attributes cost to exact fallback rather than only to the cheap floating-point stage; the retained oracle is frozen and this observation does not admit another support-search optimization. Missing stock, outer elaboration and product-provenance coverage, overlapping intervals and instrumentation prevent a scheduler-independent GPU ceiling. Peak sampled owned RSS is 15.55 GiB, with no watchdog error or termination and no owned descendant after retirement. This is a CPU diagnostic, not GPU acceleration or an independent audit.

Matched arithmetic portfolio. The complete 1,000-candidate comparison at the same two-REPL resource point shows substantial oracle benefit without established additional device gain. Native hybrid takes 1,376.24 seconds, the CPU oracle takes 769.12 seconds, and the additive GPU condition takes 762.44 seconds. Both oracle conditions preserve all 992 complete and eight rejected native outcomes and non-message fields. Their full responses agree, while 108 sources differ from native only in messages. The observed native-relative ratios are 1.789× for CPU and 1.805× for the GPU cell. Counting provider startup and complete retirement raises the GPU lifecycle to 797.61 seconds, or 1.725× relative to native and 3.70% more elapsed than CPU.

The GPU cell is only 0.87% shorter than CPU in this fixed-order shared-host observation. The device executes 488 LP models, all singleton, and wins six positive-certificate and four exactly checked negative-certificate races. These negative certificates concern feasibility of coefficient search, rather than counterexamples to Lean goals. No cross-candidate multi-model batch forms. The proxy waits for prior kernel retirement before submitting another group, while incomplete proxy inputs prevent exhaustive ready-window reconstruction. Native-ranked slowest 1% HTTP envelope sums are 449.12, 255.93 and 254.35 seconds for native, CPU and GPU. Corresponding slowest 10% sums are 1,266.11, 639.66 and 634.29 seconds. Concurrent HTTP sums are not isolated tail lifecycles. Sampled aggregate owned RSS peaks at 15.26 GiB or less, including GPU-provider host memory, with no watchdog error, termination or surviving owned process. The selected cohort met an earlier local oracle criterion.

That criterion is historical; the oracle remains a frozen baseline. These measurements establish neither a new systems contribution nor a useful extra GPU effect.

Earlier arithmetic versions, complete phase captures and the supported-expression GPU control remain in Appendix C; the supporting CPU comparisons remain in Appendix A. These observations constrain their own mechanisms and do not supply a current GPU impossibility bound.

The arithmetic CPU gains establish a stronger comparator, not a benefit from heterogeneous execution. The completed device qualifications have not established a useful additional gain over their matched CPU portfolios. Neither those observations nor the conditional cost models establish a general GPU impossibility bound.

6.5.3 Arithmetic counterexample screening

Testing intermediate goals and using counterexamples to guide backtracking are established techniques [3]; Lean also has goal-testing support [21]. Formal counterexample generation can instead produce Lean proofs of refutation [25]. Our external numeric witness check does not produce such a proof term. The question evaluated here is whether device screening reduces complete candidate cost while retaining native proof acceptance.

Arithmetic search failures can consume substantial time even when they produce no proof. Our alternative device path screens a supported arithmetic goal before `linarith` or `nlinarith` searches for a certificate. The GPU evaluates the goal and its complete supported local assumptions at many numeric points. The CPU reconstructs a proposed point as exact rational or integer values and checks every assumption and the negated goal before the tactic returns an ordinary failure. Lean can then try an enclosing proof alternative. The refutation concerns this intermediate goal, rather than the original theorem statement. Unsupported expressions, provider errors and searches without an exact witness retain the native tactic. A fully supported goal with no variables is evaluated once exactly on the CPU, avoiding repeated evaluation of the same point on the device.

A failed candidate does not establish that its intermediate goal is false. We therefore distinguish actual refuted goals from unsupported contexts and supported goals without a witness. Finite sampling under exact equalities can miss counterexamples, and such misses remain unknown. The end-to-end comparison charges source preparation, export, point construction, transfers, queueing, exact checks, fallback and process retirement. We report failed requests and timeouts separately because finding cheap counterexamples does not establish savings on the expensive failures. The paired comparison also checks accepted proof outcomes and native diagnostics separately, as required by Section 4.2. On all 7,808 miniF2F candidates, the native and device-screened paths return the same accepted set of 3,014 proofs, 4,669 Lean errors, 83 timeouts and 42 resource failures. Native execution takes 21,693.76 s and device screening 23,615.03 s, an 8.86% increase before 23.99 s of additional source preparation and native reparsing. This is one fixed-order observation on a shared host, not an estimate of run-order variance. All resource failures report allocation failure under the same 24 GiB per-process address-space limit. Peak sampled aggregate RSS is 46.87/47.38 GiB, with at most two verification processes and no aggregate-watchdog kills.

Exact checking refutes 34 intermediate goals in 29 candidates, all of which still return Lean errors. No timeout is removed. These candidates account for only 34.07 s of accumulated native HTTP envelopes; concurrent envelopes are not removable elapsed time. Actual device evaluation handles 6,187 entries in 6,186 batches, with only one cross-candidate batch. Thus this screening implementation does not improve complete verification on the tested population, and we retain it as a supporting component.

The complete cold-controller stopwatch was lost during an observation-channel outage. The detached verification client completed and all owned native processes and the device provider retired. Recorded provider lifetime also includes delayed controller recovery; it is not substituted for the missing cold performance clock. The paired cell clocks above remain independently recorded, and the missing cold clock cannot establish an additional benefit.

Answering RQ4 at its stated scope requires the remaining workload and strongest-checker comparisons. RQ4 remains unanswered until measured complete boundaries establish a device benefit or a negative opportunity bound.

6.6 Limitations

The service applies at supported native boundaries and retains the verifier as the authority. Supported backend versions and mutable runtime extensions constrain deployment. Logical equivalence alone does not promise equal resource-bounded responses. Wall-time limits depend on machine load and execution speed, so the evaluation states their operational scope separately from deterministic work limits. Valid negative and mixed results bound the claimed operating range.

Public generated attempts are replay workloads, whose arrivals can differ from a deployed multi-user service. Fixed published proof corpora remain useful controls but do not establish real candidate fan-out. Full GPU Lean checking is a distinct result from acceleration of a metadata primitive, and GPU certificate proving is a distinct workload from ordinary proof checking.

7 Discussion

Separating preparation from candidate execution connects native behavior to service resource use. Sharing preparation can reduce replication, while each candidate's mutation and lifetime determine how much of that sharing survives. The matched comparison therefore accounts for behavior, cancellation and fleet memory together, rather than treating retained context alone as the service benefit. Extending the service to another verifier requires a supported runtime boundary and qualification against its original interface.

8 Related work

Persistent verification state. Persistent Lean environments and native proof-state snapshotting retain elaborated context across attempts [13, 46]. Stock Lean also saves full elaboration or post-import snapshots [15], while Kimina provides bounded warm verification with environment reuse [41, 43]. These mechanisms provide the direct comparison for preparation reuse. The service evaluates independent native execution, cancellation and bounded fleet resources in addition to logical environment branching. This distinction requires matched evidence, rather than a novelty claim for snapshots themselves.

LeanPolish already restores native tactic state, resumes unchanged commands from saved checkpoints and supports persistent import workers [22]. Its released proof-improvement attempts include failed alternatives, while menu screening and later source verification have different boundaries [1]. These existing mechanisms strengthen the native-reuse comparison. Failure collection and saved-state execution are established baseline capabilities. LeanTree factorizes proof states according to metavariable dependencies, supporting state reuse and parallel proof search [11]. Independent proof goals and reusable intermediate states are therefore established mechanisms. A new service must demonstrate its additional behavior and resource benefit.

Axle combines preloaded Mathlib with per-request processes [50]. Its published normalized-sample comparison measures computation rather than the complete public service path. Isolated preloaded serving is therefore existing work, and our complete behavior and resource comparison with unchanged sources must establish any additional benefit.

Verification caching. Fine-grained verification caching avoids rechecking unchanged program regions [24]. The service comparison combines competent caching with warm execution, so cold startup cannot account for the entire claimed advantage. Caching and session reuse retain different parts of verification work. Native session reuse retains incremental command history and requested-artifact state [52]. Reconstruction with equivalent assertions remains a different execution history from the original native stream.

SHRED supplies exact rooted-prefix profiling and a closing-proof reuse prototype keyed by elaborated context and target, with native proof checking and fallback [47]. Reusing expensive proof construction is therefore also established work. A distinct mechanism needs evidence beyond the presence of repeated tactics or a large acceleration for a selected cache hit.

Prepared execution. Prepared execution and copy-on-write sharing are established systems mechanisms. SOCK and SEUSS share prepared state to reduce startup and increase parallel service density [40, 2]. Spice reduces disk snapshot restoration costs by co-designing the storage layout and kernel restore mechanisms [9]. A contribution beyond these prepared-execution systems requires more than specialization to verification. Our comparison tests whether native behavior and bounded shared preparation together require and benefit from a verification-specific execution mechanism.

Heterogeneous checking. GPU and CPU checker representations address different parts of verification. Nanoclo uses delayed substitutions, and nanoclo-fortran stores kernel objects in parallel arrays with integer identifiers [18, 17]. These checkers are direct competitors to representation-only acceleration. Our heterogeneous evaluation asks about the expensive dynamic checking that remains and complete verification costs, rather than extrapolating from input graph width.

Learned Lean feedback instead predicts verdicts and compiler messages on GPUs [35]. Its throughput comparison has different output semantics from native verification, and its hybrid training rewards can use predictions for unfinished checks. This is complementary evidence about an accuracy–cost tradeoff, rather than exact native checking acceleration.

Certified arithmetic and batched linear programs. Mathlib and Farkas already separate arithmetic certificate search from native proof construction [31]. Veriphy exports Lean goals and hypotheses to an external Sage oracle and checks the emitted proof, including rational LP and nonlinear arithmetic facilities [48]. These systems already cover the general export–search–check principle. The LP tactic family also reconstructs external dual coefficients into kernel-checked proofs over its supported arithmetic fragment [23]. Direct certificate reconstruction alone therefore does not distinguish our system. Exact certification of external linear programming results, including infeasibility and unboundedness, is also established [44]. Gurung and Ray assign a CUDA block to each linear program and evaluate both batch supply and complete application performance [8]. These precedents cover the numerical primi-

tives used in our heterogeneous prototypes. Their presence alone cannot establish a new verification service or predict a gain under limited concurrent request supply. Our comparison charges proof construction, exact repair, failed searches and independent cancellation at matched resources.

9 Conclusion

Native preparation and candidate execution history are separate resources in verification serving. Independent branches retain preparation while isolating candidate lifetimes, and their value depends on behavior fidelity and the equal-resource service frontier. [**RESULT: Final findings answering all four RQs, including the measured heterogeneous scope.**] The matched comparison determines whether branching supplies a systems benefit beyond existing warm reuse.

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A Matched CPU comparisons

These supporting comparisons preserve each workload, paired reference, resource configuration, native outcome exception and sampling limitation. Their separate blocks do not establish a global CPU ranking or a heterogeneous speedup.

Complete CPU compilation pairs. A separate matched four-CPU block checks every one of the 29,750 released GoedelWorkbook sources with the original GC service and the same service loading native-compiled Mathlib modules [4]. Whole elapsed time is 32,945.98 versus 15,835.68 seconds in the first pair and 38,443.59 versus 18,306.13 seconds in the reversed-order second pair. These are reductions of 51.93% and 52.38%. The third pair takes 32,484.60 versus 13,149.05 seconds, a reduction of 59.52%. The third pair retains 23 native/basic outcome changes, with ten complete outcomes becoming operational and thirteen operational outcomes becoming complete. No direct complete/rejected transition is observed. Runtime import and library loading are charged, while software build is excluded. In the first pair, native fields match on 29,725 sources after excluding the adapter timer. Sixteen operational outcomes become complete and nine complete outcomes become operational. The second pair has 21 operational-to-complete transitions and 12 reverse transitions. One additional source is operational in both but changes from a decoder failure to a timeout. The other 29,716 native payloads match except the adapter timer. No complete/rejected flip is observed, and failures remain charged. All three primary pairs finish, while the two separate resource profiles remain unfinished. These paired observations strengthen an established CPU baseline. The three observations do not establish a confidence interval or an isolated causal effect under the shared background, a new compiler contribution or GPU acceleration. Clocks from other CPU or oracle blocks are not used as denominators.

A separate producer-compatible Lean4.9 GC comparison processes all 7,808 retained miniF2F source attempts per method with and without selective native Linarith compilation [14]. All three primary pairs finish, taking 12,689.28 versus 11,723.45 seconds, 12,539.74 versus 11,192.98 seconds, and 11,914.91 versus 9,893.72 seconds. Whole elapsed reductions are 7.61%, 10.74% and 16.96%. Their descriptive mean is 11.77%, without a confidence interval. Request p99 rises from 247.85 to 275.35 seconds and from 251.14 to 253.28 seconds in the first two pairs, but falls from 224.27 to 209.72 seconds in the third. The complete count remains 3,008 in all six conditions. The second pair has two operational-to-rejected transitions and five other native diagnostic differences. The third has three such transitions and one operational diagnostic difference. Remaining payloads match except the adapter timer. Lower whole-corpus elapsed time therefore does not establish uniform tail latency, transparent native responses or GPU benefit. The two separate resource conditions also finish, completing the eight-condition acquisition. Their completion does not establish replicated memory savings. The miniF2F comparison uses a different producer version and selective compiled closure from the Goedel block above.

Multiplication-theorem application reuse. We also compare complete verification with and without reuse of resolved multiplication-theorem applications within each nonlinear arithmetic preprocessing invocation. Against the same FarkasFast/native-hybrid route, three complete 29,750-source pairs reduce whole elapsed time from 11,629.90 to 10,487.44 seconds, 12,852.84 to 11,208.51 seconds and 14,753.25 to 12,483.33 seconds. The observed reductions are 9.82%, 12.79% and 15.39%, with a descriptive mean of 12.67%. This observed range is not a confidence interval. Closed-loop request p99 also falls in each pair, from 10.56 to 10.09, 12.16 to 10.73 and 14.45 to 12.00 seconds. These are response distributions within the measured blocks, not an offered-load capacity result. The first Fast/hybrid pair also retains one operational decoder failure that becomes native rejection.

This differs from a direct complete/rejected transition and from the original-oracle product comparison below. The third pair retains six operational/completion transitions and otherwise matching native payloads. No complete/rejected transition occurs.

With the original arithmetic oracle retained in both methods, all three primary pairs take 15,385.79 versus 15,475.96 seconds, 16,245.63 versus 15,415.52 seconds, and 14,942.76 versus 12,805.48 seconds with reuse. Whole time is 0.59% longer, then 5.11% and 14.30% lower. The descriptive mean reduction is 6.28%, without a confidence interval. The third pair has five operational-to-complete and five reverse transitions. Other native payloads match except the adapter timer. No complete/rejected transition occurs. The original-oracle pairs differ in the direction of their elapsed-time changes. The different blocks and shared background do not isolate a causal oracle interaction or establish a uniform original-route gain.

Two separate Fast/hybrid resource captures complete the eight-condition comparison. Observed sampled peak PSS falls from 7,674,160 to 6,961,995 KiB, with incomplete reads in 33 of 7,051 and 34 of 6,198 non-atomic samples. These single profiles do not establish exact peaks or replicated memory savings. Seven operational/completion transitions remain. Both original-oracle resource profiles finish, completing the eight-condition comparison. Their sampled peak fleet PSS rises from 7,486,707 to 8,005,904 KiB. These single non-atomic profiles do not establish exact peaks or replicated memory savings.

These CPU observations concern an ordinary optimization within its matched block. They do not establish the globally strongest CPU method, systems novelty or GPU acceleration.

Complete native arithmetic primary series. Three matched Goedel CPU pairs compare compiled Mathlib with the same service using the native exact arithmetic oracle. Reference whole times are 10,531.82, 13,112.67 and 15,097.58 seconds. Candidate times are 11,030.02, 10,666.44 and 14,886.85 seconds. The candidate takes 4.73% longer, then 18.66% and 1.40% less time. The descriptive mean reduction of 5.11% is not a confidence interval or a stable general advantage. The direction of the timing change differs across pairs, and reference times vary even with the same code. Each condition returns all 29,750 inputs. The third pair retains 13 native outcome differences, all involving an operational outcome, with no direct complete/rejected transition. Operational failures stay charged. All eight conditions, including the two separate resource profiles, finish. In the single profile pair, observed sampled peak PSS rises from 6,561,664 to 7,478,115 KiB. Incomplete, non-atomic sampling does not establish exact or replicated peaks. This pair shows no measured memory saving. These complete comparisons preserve a competent CPU competitor. They do not establish the globally strongest CPU path or heterogeneous acceleration.

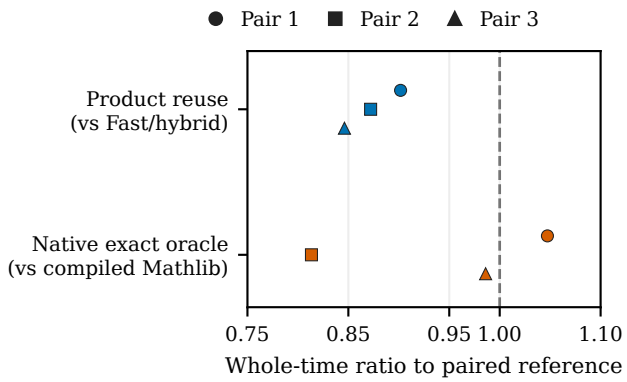


Figure 6: Whole-time ratios for two complete CPU comparisons, each processing all 29,750 GoedelWorkbook sources per condition. Each marker is one complete primary pair normalized to its own reference. The dashed line is parity. All three observations show between-pair variation, not confidence intervals. The ratio axis zooms around parity to make that variation visible. Fast/hybrid product reuse uses shared CPU eligibility 0–23, and the native exact oracle uses 0–3. Different references and eligibility prevent cross-row ranking. Separate resource profiles are excluded and operational failures remain charged. Fast/hybrid product reuse is consistently faster in these pairs, while the oracle comparison is mixed.

For the Fast/hybrid product-reuse and native exact-oracle comparisons, we divide each candidate's whole time by its own paired reference time. A ratio below one means lower elapsed time. Figure 6 displays all three primary observations per comparison. Fast/hybrid product reuse has the same direction in every pair, while the native exact oracle has both increases and reductions in elapsed time. The rows use different references and CPU eligibility, so they do not rank methods across comparisons.

Numerical support search and exact arithmetic. Three balanced primary trios compare tiered exact arithmetic, native FP64 hybrid search and HiGHS support search with exact repair over all 29,750 sources per method. They use the same native-compiled library closure, four outstanding requests and shared CPU eligibility 4–7. Relative to tiered, hybrid increases whole elapsed time by 10.61%, 5.22% and 23.43%. HiGHS takes 11.83% longer, then 11.45% less time, then 21.47% longer. These observations retain tiered as a necessary CPU competitor. The numerical paths do not establish a consistently faster replacement. They do not rank different CPU compositions. Native diagnostics also dif-

fer. One source changes from complete to rejected under HiGHS in every trio, with a recursion-depth diagnostic. All failures remain charged. All twelve conditions finish, including three separate resource profiles. Their sampled peak fleet PSS is 7,005,488 KiB for tiered, 7,668,612 KiB for hybrid and 8,723,814 KiB for HiGHS. The HiGHS capture has 1,448 incomplete samples out of 6,582. One-second non-atomic sampling and one profile per method do not establish replicated memory or CPU superiority. The recorded rusage counter includes observer, self and waited children, without validated exhaustive descendant coverage. Shared background limits causal and global CPU conclusions.

Full and restricted numerical preprocessing. Three rotated four-method blocks complete all 29,750 sources per condition, comparing Full and Fast preprocessing with native hybrid and HiGHS support search. All twelve primary conditions use shared CPU eligibility 16–19 and four outstanding HTTP requests; separate resource profiles are excluded from these primary timing comparisons. Fast changes whole elapsed time by −4.52%, +8.50% and −5.07% relative to Full with hybrid, and by +7.87%, +6.12% and −4.44% with HiGHS. It does not show a stable benefit across these observations. HiGHS takes 10.47%, 25.14% and 12.01% longer than hybrid under Full, and 24.80%, 22.39% and 12.76% longer under Fast. Basic outcomes differ for 9–15 sources per paired comparison. Each HiGHS/hybrid pair includes the same complete-to-rejected source with a recursion-depth diagnostic; operational failures and all diagnostics remain charged. These are matched composition-specific elapsed observations, not transparent replacements, confidence intervals, a global CPU ranking or GPU evidence. Shared background and sequential acquisition remain limitations.

Complete original-path cost profile. A separate instrumented capture returns all 29,750 GoedelWorkbook4.27 sources and retains body profiles for 29,689 of them. All 61 sources without a body profile have operational outcomes. The arithmetic prover `linarith` [14] and its detailed regions have the largest recorded wall-time totals for leaf regions, followed by `simp`. These thresholded trace weights measure wall intervals that can overlap across threads and requests, rather than CPU fractions or device-eligible work. The observation characterizes the original arithmetic route. Residual costs after the strongest CPU route remain to be measured.

Complete compiled-route CPU sampling. A separate instrumented compiled-route run returns all 29,750 sources and retains 5,401,763 user-CPU samples with zero reported lost samples. Exclusive sample counts and periods reconcile across flat and callchain reports. Standard `perf` children weights include callees and overlap [30]. The native `importModules` entry displays 0.36% against all recorded user-CPU period. This entry does not cover all preparation, initialization or elapsed I/O, and does not bound cold-to-warm savings. Against the same recorded user-CPU period, the displayed inclusive weights are 35.00% for the definitional-equality export in Lean's Meta layer, 16.96% for a sparse-simplex certificate-oracle entry and 10.81% for a typeclass-synthesis entry. The first symbol is exported by a `MetaM` implementation in the installed Lean4.27 source [19]. It is not a pure C++ kernel checking fraction. These entries cannot be summed as disjoint phases. Separately, reference-count decrement and small allocation account for 20.05% of native exclusive recorded period. That pair is not an exhaustive runtime category. Sampling, unknowns and callchain limits remain, and neither view supplies a decomposition of complete-verification wall time, a GPU ceiling or batching readiness. This capture characterizes the compiled original arithmetic route. Residual costs after selecting a different strongest CPU composition remain open.

Independent argument-comparison opportunity. The same sampled capture displays 23.16% inclusive CPU weight for argument comparison, including stateful work. A separate instrumented capture attributes 23.608936% of completed observed outer calling-thread CPU to entries without expression or universe metavariable flags. These quantities have different captures and scopes. Neither measures useful independent argument work or complete-verification critical-path coverage. Multiplying cross-capture fractions supplies only conditional CPU-bound sensitivity scenarios, not an end-to-end bound. The batch interface therefore has no admitted GPU provider. Another narrow backend requires evidence of useful independent work with complete setup, transfer and fallback costs.

B Historical tactic-prefix execution

These completed comparisons use their original executor and resource conditions. They do not replace the later closing-certificate service comparison in Section 6.2.

A native execution comparison tests whether these counted repetitions save complete verification time. The adapted executor uses the pinned SHRED parser, retains native proof states within each task and checks each closed root with Lean's ordinary kernel. Unsupported or failed prefixes use the original command path. The reference executes original commands in a warm native in-process batch, and both methods use four-wide native work submission. Whole clocks include parsing, preparation, transport, native checking, fallback and process retirement. This adaptation is distinct from an end-to-end execution of the published SHRED workflow. All three formal methods inherit the pinned SHRED client's or-

dinary 48 GiB verifier address-space limit. Operational resource failures and replacement costs remain in the full-population report. This comparison does not reproduce a deployed Kimina server's separate memory policy. The complete miniF2F native-batch reference returns all 7,808 sources in 28,673.00 seconds: 2,992 complete, 4,497 rejected and 319 operational outcomes. The 82 failed native requests occupy 20,672.97 seconds, or 72.10% of whole elapsed time, in non-overlapping sequential client roundtrips. This is caller waiting/roundtrip wall, not native CPU or internal tactic work. Forced retirement censors CPU accounting, and 319 source verdicts remain operationally unknown. The complete exact-text prefix capture takes 41,738.63 seconds, with 2,992 complete, 4,490 rejected and 326 operational sources. All 2,992 native-complete sources remain complete, and no new source becomes complete. Five operational sources become rejected and twelve rejected sources become operational, while 314 remain operational. The executor records seven cache lookups, 21,588 prefix nodes and 4,945 original-source fallbacks. Its included native parser cost is 10.13 seconds. The raw elapsed time is 45.57% longer than batch execution, but unresolved operational outcomes and 109 forced retirements prevent a validated speed ratio or complete CPU accounting. This does not establish savings in successful proof work or characterize the upstream closing-certificate executor. The complete unit-text prefix condition takes 40,943.77 seconds, with the same 2,992 complete, 4,490 rejected and 326 operational outcomes as exact-text prefixes. It records 3,711 cached lookups, 17,924 executed prefix nodes and 4,948 source fallbacks. All native-complete sources remain complete with a native kernel receipt, but the same twelve rejection-to-operational and five operational-to-rejection transitions preclude a validated batch-relative speed ratio. The 109 forced retirements censor CPU accounting. More cache hits therefore do not establish useful end-to-end savings on this capture. The complete original ProofNet prefix acquisition covers 5,952 sources in 186 original 32-candidate blocks. Native batch takes 1,127.28 seconds (413 complete,5,526 rejected,nine operational,four incomplete); exact prefixes take 2,502.01 seconds and unit prefixes 2,479.03 seconds (each417 complete,5,531 rejected,four operational). All413 native-complete sources remain complete with native kernel receipts. Four incomplete sources become complete; eight operational sources become rejected and three rejections become operational. These differences preclude a validated speed ratio. Exact and unit execution record 37 and 3,853 cached lookups, 13,525 and 9,716 executed nodes, and5,580/5,581 fallbacks, respectively. Their four failed native request intervals occupy 609.15/610.31 seconds; batch failure intervals occupy 13.86 seconds. These are caller roundtrip wall, not internal tactic CPU. Forced retirements censor complete CPU accounting. Reuse counters therefore do not establish useful whole-verification savings. This remains an offline executor comparison, not the complete upstream SHRED workflow or Kimina HTTP service.

Composition	Whole seconds
Compiled Fast/hybrid control	629.48
Native frontend + dual backend	475.81
Previous direct export	404.97
Natural-power direct export	350.68

Table 3: One complete original 1,000-source block with official omega, eight queues, four CPUs and a 48 GiB address-space limit. All return 992 complete and eight rejected sources. The compiled control is an adapted frontend, not pristine stock Mathlib.

C Arithmetic controls and phase observations

These supporting observations preserve earlier implementations, their own denominators, all failures and measured limitations. They are not interchangeable with the current natural-power composition.

C.1 Selected arithmetic composition

C.1.1 Arithmetic representation and service transfer

Natural-power representation. Preserving literal natural-exponent polynomial relations in the direct export reduces complete verification time on the predeclared arithmetic-heavy cohort. The previous export treated powers above two as opaque. The new representation retains those relations for exact rational expansion outside Lean, while native certificate checking remains authoritative. An admission preflight on all 111 sources with prior negative-LP/fallback evidence takes 63.96 seconds for the previous direct path and 41.30 seconds for the new path, with 109 complete and two rejected outcomes. This affected subset is separate from the subsequent complete original 1,000-source comparison.

The new path runs first, followed by previous direct export, native export with the same dual backend, and the compiled Fast/hybrid control. Table 3 gives whole times including preparation, search, checking, fallback and retirement. The ratios are 1.79502× relative to the compiled control, 1.35681× relative to the native frontend with the same dual backend, and 1.15481× relative to previous direct export. The total improvement therefore includes the numerical portfolio and direct proof path, not only exponent coverage or ordinary compilation. An exact source-body join of this complete block finds 1,888 recorded large-LP races for previous direct export and 906 for natural-power export. Counts decrease on 222 sources and increase on none. The previously identified

111-source fallback group accounts for 593 versus 176 races; the remaining 889 sources account for 1,295 versus 730. These observations support reduced pressure on this recorded path, but the guard records only problems with at least eleven rows. Smaller problems follow another CPU path; therefore the counts do not establish that total LP solves halve, attribute a causal fraction of whole elapsed time, or bound GPU benefit.

All four conditions have identical per-source verdicts and non-message fields. Relative to the compiled control, message-only differences are 108, 104 and 70; observer read errors are 55, 104, 109 and zero in acquisition order. The control uses common product-application reuse and an upstream native numerical executable. Normal output omits writing heavy diagnostic records, but the retained cost hooks still construct diagnostic JSON and read stage clocks. These are instrumented-build observations, not measurements with all diagnostic cost removed. Other CPU-eligible work shares the host.

Two subsequent complete observations alternate native–direct and direct–native order. Native/direct whole times are 615.33/348.91 and 637.27/356.54 seconds, with ratios 1.76357× and 1.78738×. All per-source verdicts and non-message fields match in each observation; each retains 108 message differences. Across the three complete observations, the ratio of whole-time sums is 1.78205×. The range is descriptive, not a confidence interval. Repetition preserves the diagnostic-cost limitation rather than eliminating it; the matched reduced-diagnostic comparison below evaluates this limitation. No GPU gain or globally fastest CPU result follows.

A subsequent same-algorithm comparison removes eager Lean diagnostic rows, stage timers and product tagging, and disables the optional process observer in both conditions. Native/direct whole times are 810.71/476.73 seconds on the same original 1,000 sources, or 1.70059×. Both retain 992 complete and eight rejected outcomes, matching all verdicts and non-message fields, with 108 message differences. This single fixed-order observation uses CPU 8–11 rather than the earlier CPU 4–7, so its absolute clocks are not pooled with the three instrumented blocks. Candidate, remote procedure call (RPC) and backend recording remain charged; the comparison is not entirely instrumentation-free and does not demonstrate GPU benefit or generalization to all Goedel sources. Direct source service spans are longer for 169 candidates, including 63 of the 70 source IDs that regress in all three earlier observations. All remain in the result. Different affinity and shared-host intervals prevent using this ID consistency to isolate diagnostic overhead or the cause of slowdown. Joining the closed numerical records to these source IDs finds no recorded large numeric race for 82 of the 169 regressions, including 20 of the 63 persistent ones. Completed HiGHS solver time across the 169 totals 1.62 seconds, excluding cancelled solves, native solvers, small paths and Lean work. This is not an all-LP cost or Amdahl bound, and does not establish the cause of these regressions.

Figure 7 shows improvement across the distribution: median service spans are 2.12–2.21 seconds for the compiled control and 1.15–1.19 seconds for direct export; p99 spans are 32.06–33.69 and 23.13–23.93 seconds, respectively. Nevertheless, 131, 164 and 156 candidates have longer direct spans in the three observations, with 70 source IDs longer in all three. No candidate is excluded on this basis. Scheduling and waiting contribute to these spans. The corresponding collected self-plus-waited-child CPU ratios are only 1.16–1.21, and that accounting is not exhaustive for all descendants. The elapsed improvement must therefore not be relabeled an equal CPU-work or energy reduction.

For the same control-ranked slowest 1% sources, overlapping service-span sums are 738.19 seconds for the compiled control, 400.42 for previous direct export and 397.37 for the new path. Some of the hardest proof tails therefore remain largely unchanged. The corresponding slowest 10% sums are 2,001.51, 1,182.14 and 1,001.16 seconds, not whole-run elapsed time. The earlier complete 29,750-source acquisition uses its frozen older build, so it does not establish generalization of this new implementation. This evidence supports an arithmetic implementation improvement, not a new solver, demonstrated GPU benefit or the complete service contribution.

Official service transfer. The complete three-method, three-block original Goedel HTTP acquisition contains 9,000 source observations. In its first two intact native/Power pairs, complete lifecycle times are 857.11/534.48 and 812.07/456.82 seconds: 1.604× and 1.778×. The ratio of summed whole times is 1.684×, and the corresponding warm ratio is 1.702×. Every cell returns the same 992 complete and eight rejected sources, with zero operational failures. Per-source verdicts and non-message fields agree; each pair retains 108 message differences. All owned descendants retire. Eight outstanding sources share four eligible CPUs 20–23 with a 48 GiB per-process limit; shared-host observations do not establish a confidence interval or globally optimal CPU execution.

The third Power condition completes in 491.13 seconds before a container interruption; recovered native execution takes 773.37 seconds. We expose this mixed-epoch 1.575× ratio separately and do not pool it as an uninterrupted third repetition. Figure 8 includes all sources. Native/Power SDK body p99 values are 49.22/33.82, 48.69/27.26 and 45.74/29.67 seconds, but 208, 152 and 245 sources have longer Power body spans. The first two native-ranked slowest ten-source sums are 993.89/650.80 and 949.66/562.37 seconds. These concurrent body spans include GC and interprocess communication (IPC) and are not removable whole time. This is service transfer of the selected CPU arithmetic composition, not GPU benefit or a new prepared-state mecha-

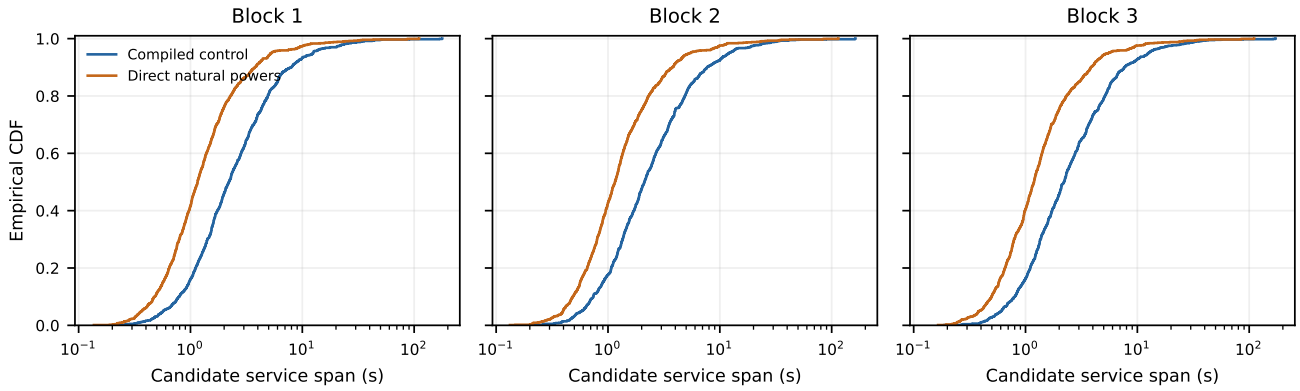


Figure 7: Candidate service-span distributions in the three complete instrumented-build observations. Each panel contains all 1,000 original candidates, including rejected inputs. Spans include waiting and overlap; they are neither isolated CPU time nor whole-run repetitions.

nism.

Existing default-engine control. A complete original 1,000-source pair also checks the reference helper's default tiered exact engine rather than assuming hybrid is globally best. The adapted compiled tiered reference takes 1,018.229 seconds and Power 492.981 seconds over the full HTTP lifecycle, a 2.065455× ratio; the warm clocks are 1,004.574 and 479.541 seconds. Both retain 992 complete and eight rejected originals, zero operational failures, and identical per-source verdicts and non-message fields; 112 message-only differences remain. All owned descendants retire. This is one fixed-order shared-host observation with eight outstanding sources and four eligible CPUs 16–19. It closes the missing default-engine control but does not replace the faster measured hybrid reference to inflate the primary gain. Different CPU epochs are not pooled, and neither global CPU optimality nor GPU acceleration follows from this alternative-reference result.

Different-prover transfer through an existing service. All 438 official DeepSeek-Prover-V2 miniF2F proof outputs finish three alternating native/Power pairs through the unchanged Kimina HTTP application and SDK. The adapted compiled reference uses the upstream helper's default tiered exact search; both use the same retained Lean 4.27 dependencies and official omega. Aggregate whole lifecycle time is 1,264.966 seconds native versus 1,194.906 seconds Power: 1.058632×, or 5.54% less elapsed. The corresponding warmed ratio is 1.061289×; individual whole ratios range from 1.049900× to 1.064890×. This range is descriptive, not a confidence interval. Each cell returns 263 valid and 175 Lean-error results, with no operational failure. Every source verdict and non-message field agrees across all six cells. One source has seven versus six warnings between methods; the full responses remain in the artifact.

Eight outstanding sources share four eligible CPUs and a 48 GiB per-process address-space limit. Whole clocks include preparation, parsing, search, native checking, fallback, queueing and owned-process retirement. The host is shared, including short root analysis/build work on those CPUs; affinity is not an exclusive resource reservation or aggregate memory bound. This is different-prover output transfer on overlapping miniF2F tasks, not a new theorem domain, an unmodified Lean 4.31 Farkas replication, or GPU acceleration. The modest gain does not extend the selected Goedel cohort's larger improvement to arbitrary Lean proofs.

C.2 Earlier device qualifications

Ready-goal device qualification. A ready-goal adapter exports arithmetic goals sequentially, solves immutable requests in parallel, then checks certificates in the same Lean state. It elaborates original arguments per goal and retains native fallback. A selected 15-source qualification forms 1,050 problems: two groups of 512 small linear problems and four smaller nonlinear groups. It checks 1,040 certificates and falls back on ten. Exported problem counts are distinct from device-ready LPs. The fresh matched native, CPU and additive GPU conditions take 116.99, 79.93 and 79.59 seconds, respectively, with seven complete and eight rejected outcomes in each. Full CPU/GPU responses agree; six message-only differences from native remain. All 56 races retain native and HiGHS competitors. Two device groups launch three models, including a two-candidate group; all retire. No GPU answer wins. The 0.42% elapsed difference in one selected preflight therefore establishes no useful GPU benefit or full-matrix admission. The first stale-library integration attempt is retained separately, rather than included as a valid additive comparison.

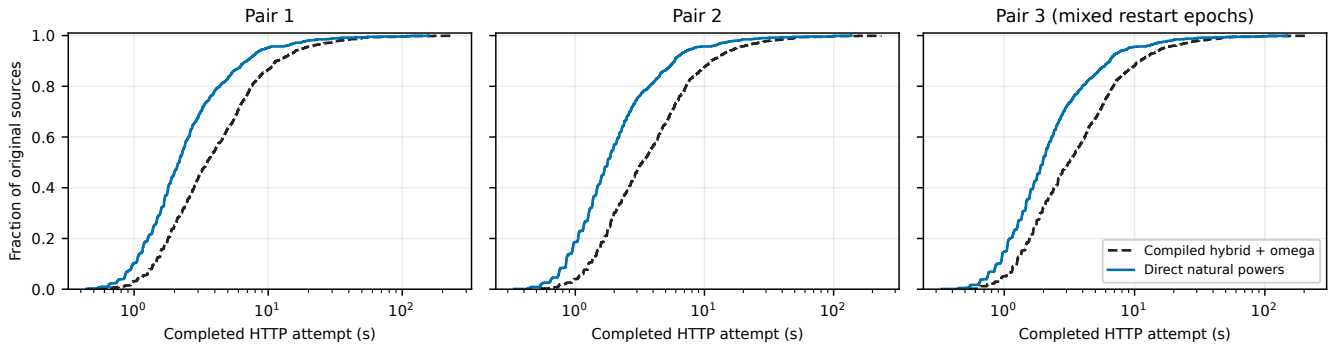


Figure 8: Completed HTTP-attempt distributions for all 1,000 original Goedel sources per method in each pair, including Lean errors. Attempt spans exclude SDK pre-dispatch waiting; whole clocks charge it. Pair 3 spans container restart epochs and is not an uninterrupted repetition.

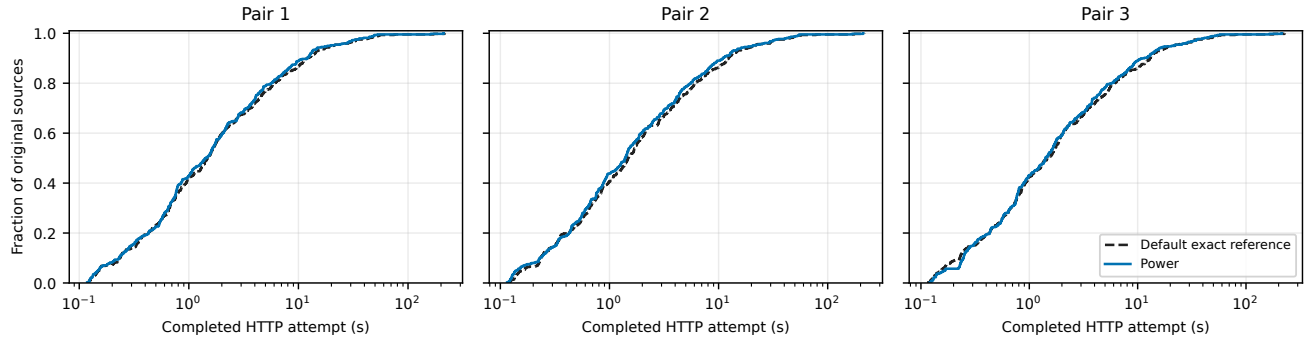


Figure 9: Completed HTTP-attempt latency for all 438 DSV2 sources in each of three alternating pairs, including the 175 Lean-error results per cell. Attempt latency excludes SDK semaphore waiting before dispatch; whole and warm sweep clocks include collective queue costs. The populations are source latencies within three paired runs, not independent whole-run trials. Neither the distributions nor overlapping body spans measure pure solver CPU or an Amdahl-removable fraction.

Independent device groups. A request-sized CUDA virtual-memory implementation uses the existing driver allocation API [36] to execute overlapping device groups under the same 48 GiB address-space ceiling. A repaired 64-source qualification executes 20 nonempty groups and 36 LP models, with complete device-timer samples, at most four overlapping groups, and no live allocations at the final provider probe. Whole warm-pipeline time is 43.07 seconds, versus 37.78 seconds for a fresh CPU oracle and 71.64 seconds for native. All verdicts and non-message fields agree. The GPU condition is 14.00% slower than CPU. This preflight does not admit a full GPU matrix. Its composition retains native hybrid but replaces the external HiGHS contender. It therefore does not establish the effect of adding GPU work while preserving the strongest measured CPU portfolio. Extra contender, transport, exact repair and retirement costs must be included in that comparison.

A corrected additive portfolio keeps native hybrid and CPU HiGHS while adding the same GPU implementation. On the original 64 sources, the same private numerical build takes 38.02 seconds without GPU and 39.33 seconds with GPU, or 3.43% more elapsed. Full CPU/GPU native responses agree. Of 68 races, HiGHS wins 46, native search 21 and an exact-checked GPU negative certificate is selected once. Six real GPU groups with nine models retire completely. These observations retain the stronger CPU competitors but do not establish a device benefit or admit a full matrix. They also show why racing cannot guarantee zero slowdown when host overhead and contention remain.

Remaining numerical work. The remaining GPU opportunity depends on numerical work in the current CPU path, including unsuccessful searches and retirement. The third two-sided CPU-oracle capture contains 1,888 uniquely source-matched numerical races. Their durations sum to 101.54 seconds, including launch, exact repair, cancellation and retirement. The 935 parsed external HiGHS searches sum to 9.25 wall seconds, while 467 additional connections are unparsed or aborted. These are not disjoint pipeline fractions. The observed critical lane, the static candidate queue finishing last, contains 14.11 seconds of complete race work.

Deleting all recorded race work while preserving other observed durations, gaps and lane assignments models a 450.47-second pipeline, or 1.53475× relative to that pair's native reference. This is an intentionally optimistic fixed-lane sensitivity, not a measured GPU result or a scheduler-independent Amdahl bound. Current product and small-LP regions are not exhaustively attributed by these records. It does not justify closing the GPU direction.

C.3 Arithmetic portfolio controls

Earlier two-sided certificate comparison. A two-sided CPU comparison tests whether reusing both successful and unsuccessful numerical search results avoids repeated exact search. This full-cohort comparison adds exact checking of external infeasibility certificates. A negative certificate establishes infeasibility of the coefficient-search problem, rather than rejection of the Lean source. With a fresh native reference, native, positive-only HiGHS and two-sided HiGHS take 681.24, 472.09 and 453.51 seconds. The two-sided ratio is 1.50216×, exceeding the 1.5× boundary by only 0.653 seconds of candidate elapsed time. All cells retain 992 complete and eight rejected sources with matching verdicts and non-message fields. Both HiGHS conditions have the same 104 message-only differences. Of 1,888 two-sided races, 334 return an exact-verified external negative certificate. This avoids some expensive exact no-certificate searches while preserving exact fallback when numerical reconstruction fails. It is a CPU observation, not GPU acceleration. The success-only and two-sided cells have 147 and 107 observer read errors, respectively. Shared background, a single fixed acquisition order and the small threshold margin preclude a stable threshold claim. An unchanged reversed-order repetition takes 687.26 seconds for native and 447.31 seconds for the two-sided oracle, or 1.53644×. It retains all 992 complete and eight rejected outcomes, matching non-message fields and the same 104 message-only differences. The oracle has 92 observer read errors.

The third native-first pair takes 691.36 versus 464.58 seconds, or 1.48814×, below the threshold. All outcomes and non-message fields again match, with the same 104 message-only differences and 105 observer read errors in the oracle cell. Across all three predeclared pairs, native and candidate whole times sum to 2,059.87 and 1,365.40 seconds: the aggregate ratio is 1.50862×. This met the historical local continuation criterion on the enriched arithmetic-heavy cohort, while preserving the sub-threshold observation. It is not a confidence interval, full-population or globally strongest-CPU result.

A complete native-frontend control with the same two-sided numerical backend takes 518.50 seconds in this block. Direct export therefore gives only 1.11607× additional speedup, or 10.40% less elapsed time. The native-frontend control alone is 1.33337× faster than the original Fast/hybrid reference. It retains all verdicts and non-message fields, with 62 message-only changes and 88 observer read errors. Frontend choice also changes LP inputs, proofs and fallback, so this is an end-to-end component comparison rather than a pure proof-construction timer. The aggregate improvement over the original reference must not be attributed entirely to the direct frontend or called a global CPU-frontier advantage. Generality beyond the selected arithmetic-heavy cohort remains open. The established certificate primitives do not alone establish the systems contribution.

A stronger arithmetic composition adds the existing native integer solver omega to both frontends. One complete matched 1,000-source block takes 454.82 seconds with native export, 388.17 seconds with direct export, and 450.99 seconds with direct export but without omega. Direct export is 1.17172× faster than the common-omega native path; adding omega to the direct path gives 1.16186×. Each condition retains all 992 complete and eight rejected outcomes with no non-message mismatch. Native versus direct has 73 message-only differences; observer errors are 80, 110 and 90 in acquisition order. One fixed-order block establishes neither confidence nor a globally optimal CPU frontier. Stock solver selection is prior art. This control narrows the incremental frontend advantage and prevents reading the earlier 1.50862× ratio as an improvement over this stronger native composition.

Common-omega phase accounting. A separate diagnostic of the earlier direct representation, before literal-power export, holds omega and the two-sided backend common to both frontends. It completes all 1,000 sources in 475.97 seconds with native export and 406.46 seconds with direct export, each with 992 complete and eight rejected outcomes. Verdicts and non-message fields agree; 65 message-only differences and 94/92 observer read errors remain. These instrumented clocks are not added to the primary timing pairs. This native frontend already includes our numerical oracle: it is a component control, not the original stock denominator.

PID-specific child resource receipts retain cancelled work. The direct cell records 33.69 and 36.14 CPU seconds of completed floating and exact LP solving, 3.22 seconds of external product generation, and 59.69 seconds in cancelled children, overlapping the completed phase totals. Unassigned child CPU is 23.36 seconds. HiGHS uses 39.03 process CPU seconds; its unassigned internal threads, startup and bookkeeping are not declared pure LP. Direct Lean certificate checks still use 183.17 calling-thread CPU seconds; native fallback product proofs use 66.81 seconds.

An explicitly conditional four-CPU capacity model excludes collector work, replaces sampled leaf CPU by exact child totals, and credits all numerical solving, external products, unknown child CPU and the entire

HiGHS process to GPU-removable work. Remaining direct CPU work is at least 1,303.41 seconds, or 325.85 seconds at ideal capacity: at most1.46068× relative to this native component control. Deleting all Lean fallback product proofs as well raises the favorable ratio to1.53960×. Recorded exact-checked negative LP answers precede 199 direct failures; 196 have unambiguous fallback stages containing 22.72 product CPU seconds. Preserving those products under the same intermediate representation and fallback protocol gives 1.51182×. Negative answers concern the exported LP, not falsity of the Lean goal. These models hold remaining work, proof paths, representation and CPU rate fixed and retain diagnostic overhead. They neither exclude redesigned systems nor justify overall GPU closure.

Sixteen interrupted numeric log fragments are retained. Reconstruction recovers intact following records only under the captured flat schema; unfinished phase CPU remains bounded by child resource receipts. The forward logger repair was not measured in this pair. Raw data, matched tails, reconstruction and model scripts are retained in the artifact; all review here is self-review.

Earlier direct-export phase capture. A separate diagnostic of the earlier direct-export build covers all 1,000 sources and 13,230 native versus 10,251 oracle numerical requests, including failed branches. Native and CPU-oracle whole times are 705.41 and 459.32 seconds, with metadata scans and gzip I/O charged. Both retain 992 complete and eight rejected sources with matching verdicts and non-message fields and 104 message-only differences between methods. Relative to its retained CPU reference, the diagnostic changes one unused-variable warning. The instrumented observation is not added to the three primary pairs.

All direct-oracle numerical-search timers sum to 29.85 seconds on successful branches and 51.98 seconds on failed branches, with 3.72 seconds of numerical products. Native fallback adds 92.03 seconds of cumulative roundtrip time. These elapsed timers include overlapping requests, and search regions also include race bookkeeping and retirement. The 935 parsed HiGHS models sum to 9.34 seconds of solver elapsed time, with 448 additional aborted or unparsed connections. Pure cancelled-worker and exact-repair CPU remain unseparated.

The parsed floating models have median dimensions of 82 rows, 230 columns and 1,752 nonzeros. Only 93 of these models overlap another candidate's request lifetime in the observed acquisition, with at most three candidates and 0.94 seconds of overlap. Wider exported-request lifetimes include communication and waiting, so they do not establish device-ready batch supply. Queueing or scheduling changes must pay their own costs.

Deleting all recorded Lean product/LP stages and complete numeric roundtrips occupies 85.81% of the native timeline and 58.76% of the CPU-oracle timeline. At fixed source paths, other durations, gaps and static queues, these optimistic deletions model native-relative ratios of 1.57600× and 1.75786×, respectively. The envelope also deletes non-GPU work, including product proofs, communication, repair and retirement. Neither model is a strict scheduler-independent Amdahl bound or a GPU result, and neither establishes the below-1.5× closure condition.

For the same sources ranked as the native slowest 1%, summed source-service spans change from 1,044.86 to 790.86 seconds. They overlap and are not whole-cohort elapsed times. On these sources, the CPU oracle's direct numerical-search timers sum to only 1.20 seconds, while recorded fallback preprocessing and linearization consume 101.27 calling-thread CPU seconds and direct plus native certificate checks consume 56.09 CPU seconds. Direct AST export is not included in those fallback-stage counters. The corresponding slowest-10% source-span sums are 2,319.47 and 1,484.35 seconds. The remaining heavy tail therefore requires examining proof and frontend work beyond the parsed floating solver.

A scheduling qualification using original source proofs uses the same private native binary and verification body for eight static lanes or eight FIFO workers. On the original first 64 sources, they take 54.81 and 58.32 seconds with equal four-CPU eligibility and the 48 GiB address-space ceiling. All source verdicts and full native responses agree. The FIFO candidate is not admitted for a full matrix. This preflight does not establish global scheduler optimality.

Earlier arithmetic acquisition and controls. The following comparisons describe earlier implementations and resource policies. Their clocks are not interchangeable with the current two-sided comparison above.

Bounded arithmetic mechanism comparison. We select 1,000 original GoedelWorkbook sources with the largest recorded `linarith` leaf-wall totals and execute them in original filename order. The selection uses the complete 29,750-source profile in Appendix A. Its overlapping trace weights rank an arithmetic-heavy opportunity cohort, rather than measuring CPU fractions or a representative population speedup.

The primary comparison uses native-compiled Mathlib and warm Kimina in-process batches of four original candidates. Its CPU reference combines Fast/hybrid with invocation-local multiplication-theorem application reuse. The prototype in Section 5.2 restores that same path on unsupported or unsuccessful direct attempts. Three whole-cohort paired repetitions alternate method order on the same permitted CPUs. The three CPU primary pairs inherit an unlimited address-space policy in both methods. The post-fault GPU and CPU-support comparison instead applies 48 GiB uniformly and retains resource failures. Clocks remain within their own measurement block, without mixing denominators

across these policies. Whole elapsed time includes preparation, export, external search, queueing, checking, fallback, capture and normal process retirement. Build time is excluded and versioned. Source identifiers and order must match the frozen input before a timing comparison is interpreted.

A predeclared threshold requires more than 1.5× complete-verification speedup with matching native outcomes. For the CPU primary series, the ratio divides aggregate reference whole time by aggregate prototype whole time across all three pairs. If the prototype passes against Fast/hybrid, the remaining credible native arithmetic competitors determine whether it also beats the fastest valid baseline. An unsuccessful executable path or missing responses do not constitute a measured negative performance result. A completed loss limits this mechanism and cohort, rather than proving a general GPU impossibility.

A separate diagnostic build attributes calling-thread CPU to preprocessing and linearization, nonlinear product generation, LP invocation and certificate reconstruction. The LP calling-thread counter excludes external solver CPU. Unattributed work remains separate, and concurrent invocation-wall sums are not divided by whole elapsed time as an acceleration bound.

Three complete paired observations give ratios of 1.347, 1.347 and 1.346 in Table 4. Each method returns all 1,000 source verdicts with 992 complete, 8 rejected and no operational failures. Full native JSON matches across repeats of each method. Between methods, the same 88 accepted sources differ solely in unused-variable warnings and their positions or counts. Raw responses remain unchanged, so outcome agreement does not establish diagnostic transparency. The ratio of aggregate complete times is 1.34668, a 25.74% elapsed-time reduction. This earlier CPU implementation finishes below the 1.5× target. The subsequent CPU/GPU comparison is complete and reported above. This earlier selected-cohort criterion is historical; the oracle is now frozen as a baseline rather than an active contribution. These CPU-only results establish neither GPU acceleration nor a general GPU impossibility.

The GPU follow-up changes external support search using cuOpt primal-dual hybrid-gradient LP search (PDLP) [38], while retaining CPU exact repair and Lean proof reconstruction. Its measured variant adds positive and negative artificial slack to each equality row and minimizes total slack, making the numerical problem feasible. The floating solution supplies an untrusted support. Already-waiting independent requests form a block-diagonal problem without a batching delay. CPU code repairs and verifies a coefficient certificate from the floating support before Lean reconstructs and checks the proof. A miss uses native fallback. Whole clocks include the host proxy lifecycle and requests to a warm numeric server. We report cold server provisioning and image loading separately from this warm-service comparison.

Header composition remains a protocol qualification for these observations. The historical CPU series imports conditional profiling and oracle handlers. The first GPU integration imports both the legacy and batch arithmetic families. Lean can try the legacy elaborator after a new elaborator fails, adding repeated work beyond the intended direct attempt and native fallback. The reported clocks describe these headers. A family-only header correction requires qualification before the results can be used as the final comparison against the leanest competent CPU reference.

The first fixed-cohort GPU run attempts all 1,000 sources in 3,374.15 seconds and returns 984 complete, 8 rejected and 8 operational outcomes. The two expired four-source request waves prevent matching verification outcomes, so this run supplies no qualified acceleration result. Feasibility alone does not ensure prompt convergence. One numerical job takes about 444 seconds and completes 296 seconds after its owner wave expires. The corresponding support request has 26 rows, 96 columns and 500 nonzeros, before artificial slack is added. The completed requests form 1,835 GPU batches from 1,854 support calls, with 1,816 singletons and 19 pairs. This serving policy yields little multirequest batching on the measured demand. The native reference under the same plan completes in 2,331.39 seconds and the robust CPU support variant in 2,363.00 seconds, both with 992 complete and 8 rejected outcomes. The GPU/reference ratio is 0.691 and the GPU fails outcome parity. These observations remain specific to the stacked headers; the corrected comparison and final best-CPU-or-GPU threshold decision remain open. All three corrected-family CPU pairs under 48 GiB are complete. Native clocks are 2,234.69, 2,234.14 and 2,222.77 seconds; direct-oracle clocks are 1,717.80, 1,751.17 and 1,695.88 seconds. The aggregate ratio is 1.29560× (6,691.60/5,164.85 seconds). Every cell returns 992 complete and 8 rejected sources without operational outcomes, with the same 88 warning-only changes. These clocks remain separate from the historical unbounded-address-space pairs in Table 4. A family-only header and owner-cancellation qualification subsequently returns 16 complete and 8 operational outcomes on 24 real sources in 630.37 seconds, against 46.05 seconds and 24 complete outcomes for the corrected native reference. Its 106 support calls produce 104 completed singleton GPU batches and two expired-owner cancellations. All observed server threads share CPUs 0–3 with that preflight's client. This variant fails native qualification and is not admitted to another full GPU comparison. The corrected CPU paired series is complete, while heterogeneous execution remains open. A Windows archive-argument overflow loses the preflight server stdout and exact provisioning/retirement stopwatches; client models, numeric replies, batch events and native responses remain retained. The cached-container start-to-killing event span is 642 seconds, not an exact cold lifecycle measurement. The temporary target is re-

Whole cohort	Pair 1	Pair 2	Pair 3
Fast/hybrid CPU (s)	2,282.741	2,364.280	2,325.932
Direct oracle CPU (s)	1,694.375	1,755.730	1,727.780
Reference / prototype	1.347	1.347	1.346

Table 4: Complete native verification of the same enriched 1,000-source cohort per cell, with alternating method order. Whole times include the common resource observer and normal retirement. These historical, unbounded-address-space CPU-stage pairs fall below 1.5×; the final CPU/GPU comparison remains open. They establish no GPU benefit.

Native diagnostic region	Recorded CPU (s)
Preprocessing and linearization	331.667
Nonlinear product proof generation	288.030
Lean LP caller	9.837
External numerical LP search	979.803
Certificate construction/checking	275.211

Table 5: Complete selected 1,000-source native diagnostic. Calling-thread CPU in distinct processes is separate from elapsed time, and these counters are not an exhaustive whole-pipeline CPU partition.

moved and the other Workspace remains running.

In the first pair, the common resource observer consumes 312.54 and 224.17 observed collector CPU seconds in the two methods. Whole clocks include that observer, rather than measuring an observer-free verifier effect. The recorded collector and waited-child counters are not exhaustive. Deduplicated sampled descendant self-CPU maxima, plus collector CPU, give lower bounds of 3,368.36 and 2,577.75 CPU seconds for the two methods, above their recorded totals of 2,387.68 and 1,941.90 seconds. Sampling misses final tails and unseen processes, so these bounds neither determine exact CPU reductions nor replace the complete wall clocks. A four-source Fast/hybrid control takes 55.060 seconds with the observer and 54.559 seconds without it, with identical native responses. This single fixed-order diagnostic does not correct the 1,000-source clocks or quantify method-specific observer effects. The first reference uses the same executed Fast/product code as later pairs, but predates a repair to the then-unused prototype fallback in the common library. That build lineage remains explicit. The interrupted third attempt has no recoverable complete raw result and is excluded. Its replacement uses rebuilt pinned dependencies after source-native qualification; every same-method native response matches the earlier repeats. Sampled baseline peak PSS is 8.725, 4.656 and 9.986 GB across the three pairs, while prototype peaks are 4.345, 4.355 and 9.127 GB. This variation and non-atomic one-second sampling do not establish stable or halved memory savings. The complete all-call diagnostic separates numerical search from native proof work without treating overlapping wall intervals as a disjoint partition. It returns the same 992 complete and 8 rejected native responses as the primary reference, with no operational outcome. All 13,414 numeric requests are retained. Table 5 reports calling-thread CPU for preprocessing, nonlinear product proof generation, the Lean LP caller, external numerical search and certificate construction. Final source validation, collector work and other uninstrumented operations remain charged by whole elapsed time.

External LP search occupies 42.28% of the 2,319.668-second diagnostic timeline. This phase-specific interval union is occupancy, not removable critical-path time. The Lean LP interval also includes external waiting, so adding their wall intervals would count the same wait twice. The direct-oracle diagnostic retains all 10,257 numeric requests, including failed direct searches and native fallbacks. Its LP search consumes 627.840 CPU seconds and numeric product generation 3.372 CPU seconds. The numeric-work interval union occupies 35.25% of its 1,744.056-second timeline. Freezing certificates, fallback choices and all other work while removing the 631.870-second numeric interval sum gives a conditional 1,112.186-second sensitivity. Changing solver supports can change proof and fallback work, so this calculation is neither achieved acceleration nor a certified end-to-end bound.

A separate numeric-only replay accounts for complete worker CPU through actual exit. Replaying the native baseline's 13,414 requests and two handshakes finds no certificate or error difference and consumes 1,027.770 CPU seconds. LP search accounts for 1,018.047 seconds, comprising 40.015 FP64 and 978.032 exact-search seconds. Request parsing and response packing/writing consume 4.567 seconds, LP construction 0.503, exact repair 0.369 and exact verification 0.074. The remaining 4.209 seconds include diagnostic writes, input reading, process startup/exit and other worker work. This complete numeric-service composition does not retroactively partition an earlier sampled 976-second service observation or replace Lean end-to-end wall time.

Every native request records its matrix shape, with median 10 rows, 20 columns and 74 nonzeros. The 99th percentiles are 183, 552 and 4,648, respectively. Four requests lack individual source labels, so global costs retain them while candidate-specific attribution remains bounded. Generated-product ancestry was not recorded for this native request stream and remains unknown rather than zero. The 288 seconds of Lean product proof construction and 3.372 seconds of direct numeric product generation represent different work. The measured GPU LP backend does not replace native product proof construction.

The full prototype diagnostic preserves every native response on the same 1,000 sources. It records 3,597 successfully checked contradiction branches and 2,953 fall-

back attempts, with no malformed or unmatched telemetry. Of the 1,000 distinct source bodies, 759 have an observed direct check, 273 have a fallback and 32 have both. Thus branch/fallback record counts are not a mutually exclusive tactic success rate. Failure labels are 2,348 unsupported-carrier and 605 expected-array records; they remain unnormalized rather than assigning an unsupported solver-failure cause.

On the observed successful branches, export sums to 0.48 seconds, the search roundtrip to 379.95 seconds and certificate checking to 233.33 seconds. Inside that roundtrip, native Rust linearization, numeric products and LP search sum to 0.29, 2.47 and 151.09 seconds respectively. Parent and nested timers must not be added. Median potential pools contain 136 facts, while certificates use a median of four terms; the observed maxima are 946 and 84. These successful-path wall sums neither exhaust failed/unsupported work nor give whole-verification CPU shares or a GPU upper bound. All failed work and fallbacks are still charged by the complete primary clocks. The numeric product work already removed from Lean is small on these observed successful branches. No additional GPU measurement or new kernel is inferred from it.

Complete matched queue-family wall-cost profile. A complete diagnostic repeats the same 1,000 sources, corrected family headers, upstream eight sequential queues, four CPUs 0–3 and 48 GiB allowance. Its own native and CPU-oracle clocks are 691.20 and 545.05 seconds. Each returns 992 complete and eight rejected sources without operational outcomes; every native response JSON value exactly matches its corresponding noninstrumented primary. These clocks remain separate from the primary 685.73/537.31-second pair and earlier wave-only captures.

All 13,230 native and 10,251 CPU-oracle numerical calls carry source, invocation and matrix/product metadata, including failed searches and fallbacks. The CPU oracle has 3,597 checked direct certificates, 605 direct failures, 5,440 fallback certificates and 609 fallback failures. The four observed native arithmetic regions appear in Table 6. Wall occupancies overlap across concurrent candidates. Other frontend, export, kernel, observation and retirement work remains charged to the whole clock. Ring, defeq and typeclass work stays inside the corresponding Lean preprocessing and proof-construction regions; product timing includes building Lean proofs.

The CPU-oracle numerical services consume 554.83 CPU seconds: 546.83 in floating-point and exact LP search, 3.13 in numeric product generation, 0.26 in exact repair and 0.04 in exact coefficient verification. Parsing, packing, linearization, response and dispatch account for the remainder. Rust phase CPU is made exclusive for nested timers; raw wall intervals retain their observed overlap. Failed direct branches alone consume 76.60 LP CPU seconds and failed fallback branches 44.28 seconds.

Native arithmetic region	CPU (s)	Wall occupied (%)
Hypothesis preprocessing/linearization	250.70	52.40
Nonlinear product proof construction	211.56	48.50
Exported numeric FP/exact LP search	859.53	78.77
Lean certificate construction/checking	208.57	47.13

Table 6: Complete native eight-queue diagnostic on 1,000 sources. CPU counters come from the relevant Lean calling threads or numerical children. Each wall occupancy is the region's interval union divided by the diagnostic's own 691.20-second clock. Regions overlap and neither column is a disjoint whole-wall partition.

All exported numeric products and LP search sum to 833.99 interval seconds on the 545.05-second oracle clock. Overlap makes this exceed whole elapsed time, so subtracting that sum provides no useful finite ceiling. Their union is 411.68 seconds, 75.53% of that same clock. An optimistic fixed-timeline compression deleting every occupied instant leaves 133.37 seconds and gives $691.20/133.37 = 5.18274\times$ against the matched diagnostic native reference. This also deletes concurrent non-GPU work. It is neither a scheduler-invariant bound nor measured GPU acceleration. Neither calculation establishes a below-1.5× ceiling.

The cheap floating search fails on 1,888 oracle calls, which then consume 518.39 CPU seconds in exact search. Their median dimensions are 82 rows and 230 columns. Exact searches from at least two distinct candidates overlap for 221.40 seconds. These are observed CPU-search availability intervals; a GPU still owes model preparation, transfer, queueing, repair and cancellation.

Type coverage also matters. The direct adapter accepts real and rational carriers; 2,348 unsupported-carrier fallback events occur on ten natural- or integer-source candidates. Native fallback processes those sources in the same cohort. Event counts include recursive tactic branches and are not distinct proof counts or predicted savings.

Figure 10 reports the native LP-size distribution. The retained 11-row gate was chosen from an earlier complete native capture and is unchanged here. The new capture confirms that the 7,201 smaller calls account for only 0.02463 seconds of numeric LP-search CPU. The other 6,029 calls contain almost all numeric search work.

Concurrent native and heterogeneous comparison. With eight-item waves and the same CPUs 0–3, the pooled native reference reuses independent native numerical-daemon handles with no LP search under the handle-list mutex. It takes 1,237.37 seconds for all 1,000 sources. The CPU AST oracle takes 1,037.85 seconds, a

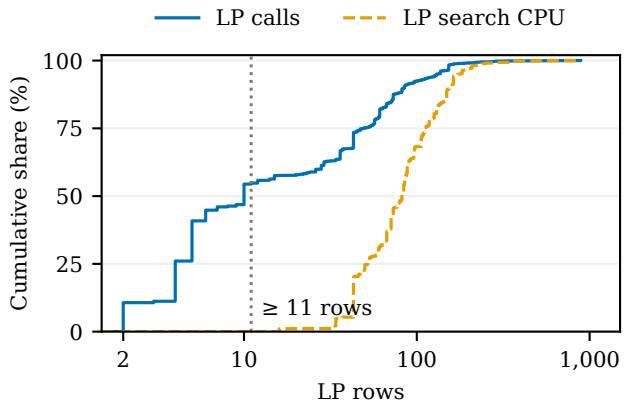


Figure 10: LP-size distribution in one complete pooled native diagnostic (13,230 calls). The solid curve weights calls equally; the dashed curve weights floating-point and exact LP-search CPU. Calls below the retained 11-row gate are 54.43% of calls but only 0.00287% of LP-search CPU. This single-capture distribution is neither a confidence interval nor a GPU speedup estimate.

1.19224× ratio. Both return 992 complete and 8 rejected sources, with matching verdicts and non-message native responses; 88 source payloads differ only in warnings. A serial-native control takes 1,600.63 seconds and is not used as the strongest reference. These observations form one complete pair, without replicated estimates. The width-four diagnostic above does not provide a zero-cost bound for this faster pooled reference.

The GPU implementation races each eligible LP against the original CPU hybrid; only an exactly repaired coefficient certificate can win. It cancels and reaps the other numerical child, and batches at least two live requests from concurrent candidates. Withdrawal cancels a shared job and requeues its surviving members while their CPU shadows continue. The full matched 1,000-source GPU cell takes 1,020.12 seconds with a prepared upstream server, or 1.21296× relative to the pooled native reference. Its complete cold lifecycle takes 1,036.25 seconds, giving 1.19408×. The warm observation is 1.71% below the CPU oracle; the cold lifecycle is only 0.15% below it. This is one full observation, not replicated evidence of a stable GPU advantage, and does not meet the 1.5× target. Every source verdict and non-message native field matches; the GPU has 98 warning-only response changes and zero operational failures.

The GPU submits 384 actual cross-candidate jobs: 358 pairs, 20 triples, four groups of four and two groups of five. Of these, 311 are cancelled after member withdrawal and 73 return results; all 384 retire. There are 4,656 labelled numerical races, with 3,352 CPU-certificate wins, 1,214 CPU-no-certificate completions and 90 exactly repaired GPU wins. Cancellation can therefore consume batch work even when the native verification result remains unchanged. Completed jobs report 18.399 server solver seconds; device compute for cancelled jobs is unknown and is not counted as zero. The 83.90 seconds of summed client batch/retirement intervals include RPC, overlap and cancellation, rather than measuring device-kernel time.

For the same ten sources selected as the slowest 1% on the native reference, summed command-service spans are 505.97 seconds native, 389.99 seconds CPU oracle and 391.76 seconds GPU. For the slowest 10%, the corresponding sums are 1,266.45, 948.40 and 927.69 seconds. These spans overlap and are not whole-cohort clocks. GPU does not improve the CPU oracle on the 1% group. GPU resource sampling has 203 process-read errors during child exit; its sampled memory peaks exclude the separate server and support no resource-superiority claim.

Child creation, communication, contention and cancellation can make a race slower than its CPU shadow. The design alone supplies no worst-case no-slowdown guarantee. The paired cost capture above uses its own clocks. The eight-item-wave submission here is not a continuously replenished serving frontier; queue scheduling remains a separate control for the common baseline. A full known-answer diagnostic on the same 1,000 sources has a fresh native clock of 1,228.00 seconds. Returning actual recorded coefficient answers while retaining Lean checking takes 866.84 seconds through the native path and 767.62 seconds through the AST-oracle path (1.417× and 1.600×, respectively). Every cell returns 992 complete and eight rejected sources with identical verdicts and non-message payloads; all 13,230 native and 10,251 AST numerical calls hit the recorded answers. The AST control has 88 accepted-source warning-only changes. This answer-aware diagnostic is excluded from valid CPU/GPU speedup scores. It retains process setup, lookup and IPC and is not a strict zero-cost upper bound. Its result therefore neither establishes achievable GPU acceleration nor supports a below-1.5× impossibility claim. A subsequent native-only cell uses the upstream eight sequential queues on the same 1,000 sources, at most eight candidate tasks, four CPUs 0–3 and 48 GiB. It completes in 685.73 seconds with identical complete native response JSON to the earlier wave reference. Its matched CPU-oracle cell completes in 537.31 seconds (1.27622×). Both conditions return 992 complete and eight rejected sources with matching verdicts and non-message payloads; the oracle has 88 accepted-source warning-only changes. This is one complete pair, not replicated evidence. The wave GPU ratio and answer-aware control above cannot be transferred to this stronger CPU submission interface. The upstream queue API is a baseline correction, not our systems contribution. Negative route closure requires applicable evidence that GPU cannot push the best valid overall result beyond 1.5×.

A fresh full-cohort comparison adds HiGHS support search with the same exact CPU shadow and native

CPU comparison	Whole elapsed versus its own paired reference
GC/native libraries, Goedel	51.93%,52.38%,59.52% lower; separate profiles unfinished
Selective native arithmetic, miniF2F	7.61%,10.74%,16.96% lower; mixed p99
Fast/hybrid product reuse	9.82%,12.79%,15.39% lower
Original-oracle product reuse	0.59% longer;5.11%,14.30% lower
Native exact oracle	4.73% longer;18.66%,1.40% lower
FP64 hybrid over tiered	10.61%,5.22%,23.43% longer
HiGHS over tiered	11.83% longer;11.45% lower;21.47% longer

Table 7: Supporting CPU comparisons, with each effect tied to its own matched block. Complete source populations, resource eligibility, failures and native response exceptions remain in Appendix A. Between-run variation is descriptive, not a confidence interval or cross-row ranking.

eight-queue interface. Native, CPU-only and HiGHS paths take 691.10, 529.17 and 462.50 seconds, respectively. The best observed ratio is 1.49425×, short of 1.5× by 1.77 seconds in this single pair. Each returns 992 complete and eight rejected sources; verdicts and non-message response fields agree, with 104 warning-only source changes for HiGHS. The HiGHS cell has 150 process-observer read errors, so its sampled memory peak is incomplete. Native-selected slowest 10% task spans sum to 2,313.71 versus 1,502.73 seconds; their occupied unions are 656.44 and 439.78 seconds, not independent-subset speedups. This stronger published solver is a competing CPU mechanism, not our systems novelty. Its LP calls sum to only 7.30 seconds of overlapping wall time, while 1,214 exact no-certificate race decisions sum to 177.76 seconds. Neither sum is a disjoint whole-wall share. The result motivates examining the cost of establishing failed searches as well as successful supports; it does not establish a below-1.5× GPU ceiling or repeated threshold performance.

Matched CPU competitors and controls. Table 7 summarizes the complete primary observations used to qualify CPU alternatives. Each comparison uses its own version, paired reference and CPU eligibility. Full clocks, native outcome changes, resource observations and sampling limitations remain in Appendix A. These supporting controls strengthen the CPU baselines used for comparison. They do not make ordinary compilation or invocation-local reuse the system contribution. The ProofNet measurements below use their own reference.

Whole corpus time (s)	Run 1	Run 2	Run 3
Native warm pool	310.478	311.544	318.572
Native snapshot/cache	319.524	320.059	331.712
Packed CPU control	319.510	326.852	321.050
GPU provider	315.870	322.891	323.607

Table 8: Complete source verification for 5,952 ProofNet-lineage attempts per condition. All twelve primary conditions finish with identical basic outcomes within each matched block. Four separate resource profiles also complete. No complete miniF2F GPU comparison is available.

Complete ProofNet source verification. Three complete primary comparisons process the same 5,952 retained ProofNet-lineage [42, 7] Lean4 source attempts in original order, including failed attempts. Each method uses four frontend workers on the same four permitted CPUs, the retained Lean4.9.0-rc1/Mathlib environment, default native runtime threads and the original 300-second per-attempt allowance. The native warm pool reuses prepared import headers. The native snapshot/cache condition retains full native snapshots and caches only complete results for exact matches of both import header and candidate source. The GPU method uses an RTX 5090 and includes conservative host admission, packet construction, device initialization, transfer, synchronization, CPU fallback, enclosing source verification and final native checking. Lossless archiving of experiment outputs occurs after the measured final EOF. CPU affinity does not provide exclusive cores. GPU time slicing shares the device with another Workspace.

Table 8 rejects an end-to-end acceleration claim for this supported expression reduction in Lean's metaprogramming (Meta) layer on the measured cohort. Relative to the faster native baseline in each block, GPU execution is 1.74%, 3.64% and 1.58% slower, with a mean of 2.32%. These values describe the observed repetitions, not a confidence interval. Every condition has 413 complete, 5,534 rejected, four incomplete and one retained operational outcome. The packed CPU version is a component control. It rejects unsupported operations at a different point from the GPU path, so their time difference cannot isolate device placement alone. The strongest native baseline grounds the end-to-end comparison.

Native operation/readback logs establish actual GPU engagement, while unsupported work remains native. The implemented provider launches one CUDA thread [37] for each supported beta or zeta reduction. These reductions substitute an argument into a function body or a value into a let-bound expression. The provider then synchronizes before returning to Lean. It does not batch independent reductions or execute the complete Lean kernel on the device.

Across each primary repetition, retained counters

record 2,201 supported operations out of approximately 2.38 million provider calls, corresponding to 0.0925%. The percentage measures coverage by call count, not the fraction of verification time that could be accelerated. Each GPU condition has one failed worker without terminal counters, so the totals are observed rather than exhaustive. Per-process cumulative maxima give summed provider wall times of 3.169, 4.921 and 6.158 seconds, including initialization sums of 1.326, 1.178 and 1.274 seconds. These timers include waits and overlap across workers. Subtracting them from whole-corpus makespan would not identify the cause of the slowdown.

This implementation adds host admission, packet preparation, transfers and synchronization while leaving most calls on the native path. The complete results reject an acceleration claim for this implementation on this cohort. They do not establish whether GPU sharing caused the slowdown. No matched co-tenant activity comparison isolates that effect. [**RESULT: Complete miniF2F comparison, cost-weighted supported-work coverage and remaining strongest-checker comparisons.**]

Four separately instrumented complete conditions observe whole times of 328.046 seconds for warm execution, 348.390 for snapshot/cache, 338.670 for the packed CPU control and 342.949 for GPU execution. Their sampled fleet PSS maxima are respectively 5,289,626, 9,268,751, 5,161,250 and 6,315,211 KiB. These are single instrumented conditions. One-second non-atomic sampling misses short-lived processes and peaks, and the sampling process contributes CPU overhead. These clocks do not replace the primary comparisons or establish replicated memory uncertainty. GPU context and driver memory, as well as other owners' memory, are outside host-PSS figures.